Chapter 1
Proofs by calculation

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you will learn

- Turn hypotheses into a chain that begins and ends at the required expressions.
- Separate algebraic identities from substitutions of known facts.
- Control the direction of inequalities and justify sign conditions.
- Translate individual steps into `calc`, `ring`, `rw`, `rel`, `numbers`, `extra`, and `addarith`.

A proof is an explanation

Every link must be justified, and the links must combine to establish the stated goal.

Start by identifying the number system

Domain	Examples	Lean name
$\mathbb{N}$	$0, 1, 2, \ldots$	`Nat`
$\mathbb{Z}$	$\ldots, -2, -1, 0, 1, \ldots$	`Int`
$\mathbb{Q}$	$-\frac{3}{2}, \frac{5}{7}, 4$	`Rat`
$\mathbb{R}$	$\frac{5}{7}, \sqrt{2}, \pi$	`Real`

- Here 0 is a natural number.
- The domain constrains the legal operations: $3 - 5$ in $\mathbb{N}$ is truncated subtraction.
- In calculations over $\mathbb{Q}$ or $\mathbb{R}$, division by a known nonzero constant is available.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 1 introduction and Section 1.3.

1.1 A chain proves its endpoints equal

Suppose each of the following statements has been justified:

$$E_0 = E_1, \qquad E_1 = E_2, \qquad E_2 = E_3.$$

Transitivity of equality gives $E_0 = E_3$.

A useful writing discipline

Start with the left side of the goal. End with its right side. Keep an equality sign between consecutive expressions.

Two common justifications:

- **Algebra:** the expressions agree for all values in the domain.
- **Substitution:** a hypothesis says that one expression equals another.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.1.

Worked example: reveal the expressions you know

Let $a, b \in \mathbb{Q}$, with $a - b = 3$ and $ab = 4$. Prove $(a + b)^2 = 25$.

$$\begin{aligned} (a+b)^2 &= (a-b)^2 + 4ab && \text{algebra} \\ &= 3^2 + 4 \cdot 4 && \text{the two hypotheses} \\ &= 25 && \text{arithmetic.} \end{aligned}$$

Why does the first step hold?

$(a - b)^2 + 4ab = a^2 - 2ab + b^2 + 4ab = a^2 + 2ab + b^2$.

The initial rewrite was chosen because $a - b$ and ab have known values.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.1; numerical variant of Example 1.1.1.

A calculation can avoid an unnecessary division

Let $a, b, c, d, e, f \in \mathbb{Z}$, with $ad = bc$ and $cf = de$. Then

$$\begin{aligned} d(af - be) &= (ad)f - dbe \\ &= (bc)f - dbe \\ &= b(cf) - dbe \\ &= b(de) - dbe = 0. \end{aligned}$$

Every step uses multiplication, subtraction, or a given equality.

Check the hypotheses before cancelling

Writing $a = bc/d$ would require $d \neq 0$ and a suitable domain. The hypotheses do not give $d \neq 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 1.1.4.

Practice 1.1: write the chain before simplifying

1. Let $r, s \in \mathbb{R}$, with $r + 3s = 2$ and $s = -1$. Prove $r = 5$.
2. Let $a, b \in \mathbb{Q}$, with $a - b = 5$ and $ab = 6$. Prove $(a + b)^2 = 49$.

For each proof:

- State the first and last expressions before filling the middle.
- Mark every step as algebra, substitution, or arithmetic.

Practice 1.1: solution discussion

1. Introduce the expression $r + 3s$:

$$\begin{aligned} r &= (r + 3s) - 3s && \text{algebra} \\ &= 2 - 3(-1) && \text{substitution} \\ &= 5 && \text{arithmetic.} \end{aligned}$$

2. Use the same identity as in the worked example:

$$(a + b)^2 = (a - b)^2 + 4ab = 5^2 + 4 \cdot 6 = 49.$$

The calculation proves the requested expression directly; values of a and b are unnecessary.

1.2 Read a Lean statement as a mathematical sentence

```
example {a b : Rat}
    (hd : a - b = 3) (hp : a * b = 4) :
    (a + b) ^ 2 = 25 := by
```

- `a b : Rat`: the variables are rational numbers.
- `hd` and `hp`: names for the hypotheses.
- After the final colon: the proposition to prove.
- `by`: the proof will be given by tactics.
- Multiplication uses `*`; exponentiation uses `^`.

The examples use the textbook environment: its chapter file imports `Library.Basic` and runs `math2001_init`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.2 and accompanying Lean file.

The calculation, with one justification per line

```
example {a b : Rat}
    (hd : a - b = 3) (hp : a * b = 4) :
    (a + b) ^ 2 = 25 := by
  calc
    (a + b) ^ 2 = (a - b) ^ 2 + 4 * (a * b) := by ring
    _ = 3 ^ 2 + 4 * 4 := by rw [hd, hp]
    _ = 25 := by numbers
```

- `_` repeats the preceding right side.
- `ring` proves a polynomial identity.
- `rw [hd, hp]` substitutes the named equalities.
- `numbers` checks the final numerical equality.

Source: Heather Macbeth, *The Mechanics of Proof*, Method of Example 1.2.1.

Treat Lean feedback as a local question

At a line of a `calc` proof, ask:

1. What equality must this line establish?
2. Does it hold by algebra alone, or does it use a hypothesis?
3. If it uses a hypothesis, does its left side occur in the expression?

Parentheses matter to rewriting

$(ab)c$ and $a(bc)$ are equal, but have different syntactic structure. Use an algebraic step to expose the exact subexpression you plan to rewrite.

An unfinished placeholder is still an unfinished proof. Check that all proof obligations are resolved.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 1.2.2–1.2.4.

Practice 1.2: select the justification

Let $r, s \in \mathbb{R}$, with named hypotheses

$$\texttt{h1} : r + 3s = 2, \qquad \texttt{h2} : s = -1.$$

Fill the right column with `ring`, `rw [...]`, or `numbers`.

Step	Justification
$r = (r + 3s) - 3s$	_____
$(r + 3s) - 3s = 2 - 3s$	_____
$2 - 3s = 2 - 3(-1)$	_____
$2 - 3(-1) = 5$	_____

Which step would fail if you tried `ring` without using any hypothesis?

Practice 1.2: a complete Lean solution

```
example {r s : Real}
    (h1 : r + 3 * s = 2) (h2 : s = -1) : r = 5 := by
  calc
    r = (r + 3 * s) - 3 * s := by ring
    _ = 2 - 3 * s := by rw [h1]
    _ = 2 - 3 * (-1) := by rw [h2]
    _ = 5 := by numbers
```

`ring` cannot establish the substitution steps for arbitrary r, s. The hypotheses supply their mathematical content.

1.3 How to find the intermediate expressions

1. Write the target endpoints.
2. List useful expressions appearing in the hypotheses.
3. Rewrite the target to contain those expressions.
4. Substitute only after the matching expressions are visible.

A small example

If $x + 6 = 2$ in $\mathbb{Z}$, then

$$x = (x + 6) - 6 = 2 - 6 = -4.$$

Adding and subtracting 6 creates the expression we know.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.3; method of Example 1.3.2.

Worked example: combine two equations

Let $x, y \in \mathbb{Q}$, with $x + 2y = 9$ and $3x - y = 13$. Prove $x = 5$.

Planning

The coefficients of y are 2 and -1. Adding the first expression to twice the second cancels y.

$$\begin{aligned} x &= \frac{(x + 2y) + 2(3x - y)}{7} && \text{algebra} \\ &= \frac{9 + 2 \cdot 13}{7} && \text{substitution} \\ &= 5 && \text{arithmetic.} \end{aligned}$$

The denominator 7 is nonzero in $\mathbb{Q}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.3; methods of Examples 1.3.6–1.3.8.

Worked example: organize a polynomial around a hypothesis

Let $w, z \in \mathbb{R}$ and suppose $w + 2 = z$. Prove

$$w^2 + 5w + 4 = z^2 + z - 2.$$

$$\begin{aligned} w^2 + 5w + 4 &= (w+2)^2 + (w+2) - 2 && \text{algebra} \\ &= z^2 + z - 2 && \text{substitution.} \end{aligned}$$

Check the rewrite:

$$(w+2)^2 + (w+2) - 2 = w^2 + 4w + 4 + w + 2 - 2 = w^2 + 5w + 4.$$

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.3; method of Example 1.3.9.

Practice 1.3: create the expressions you need

1. If $x + y = 7$ and $2x - y = 2$ in $\mathbb{Q}$, prove $x = 3$.
2. If $a - 2 = b$ in $\mathbb{R}$, prove $a^2 - a + 1 = b^2 + 3b + 3$.

For the first problem, find a linear combination that cancels y.
For the second, rewrite the left side using $a - 2$.

Practice 1.3: solution discussion

1. The y coefficients already cancel:

$$x = \frac{(x+y)+(2x-y)}{3} = \frac{7+2}{3} = 3.$$

2. Complete a polynomial in $a-2$:

$$\begin{aligned} a^2 - a + 1 &= (a-2)^2 + 3(a-2) + 3 \\ &= b^2 + 3b + 3. \end{aligned}$$

Both proofs follow the same pattern: algebra first, substitution second.

1.4 An inequality needs a direction and a sign condition

For real numbers:

Known fact	Additional condition	Valid conclusion
$a \leq b$	none	$a + c \leq b + c$
$a \leq b$	none	$-b \leq -a$
$a \leq b$	$c \geq 0$	$ca \leq cb$
$a < b$	$c > 0$	$ca < cb$
$a \leq b$	$0 \leq a$	$a^2 \leq b^2$

Transitivity

$A \leq B < C \leq D$ implies $A < D$. A consistent chain with a strict step has a strict conclusion.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4.

Worked example: subtraction reverses the comparison

Let $x, y \in \mathbb{Z}$, with $x + 4 \leq 2$ and $y + 3x \geq 1$. Prove $y > 6$.

$$
\begin{aligned}
y &= (y + 3x) - 3x && \text{algebra} \\
&\geq 1 - 3x && y + 3x \geq 1 \\
&= 13 - 3(x + 4) && \text{algebra} \\
&\geq 13 - 3 \cdot 2 && x + 4 \leq 2 \\
&= 7 > 6 && \text{arithmetic.}
\end{aligned}
$$

When $x + 4$ becomes larger, $13 - 3(x + 4)$ becomes smaller. That is why the fourth line points toward $\geq$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4; method of Example 1.4.1.

Worked example: add a nonnegative square

Let $x, y \in \mathbb{R}$ and suppose $x^2 + y^2 \leq 2$. Prove $(x+y)^2 < 5$.

$$
\begin{aligned}
(x+y)^2 &\leq (x+y)^2 + (x-y)^2 && (x-y)^2 \geq 0 \\
&= 2(x^2+y^2) && \text{algebra} \\
&\leq 4 && \text{the hypothesis} \\
&< 5 && \text{arithmetic.}
\end{aligned}
$$

How was the extra square chosen?

Its expansion cancels the $2xy$ term and produces the known expression $x^2 + y^2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4; method of Example 1.4.8.

Valid individual comparisons can form an invalid proof

If $n \in \mathbb{Z}$ and $n \geq 6$, all these statements are true:

$$n^2 \geq 36 > 28 \leq 3n + 10.$$

They do **not** combine to prove $n^2 > 3n + 10$: the direction changes. For comparison, $4 > 3 \leq 100$ does not imply $4 > 100$.

A correct chain

Since $n \geq 6 > 0$,

$$n^2 = n \cdot n \geq 6n = 3n + 3n \geq 3n + 18 > 3n + 10.$$

Keep some dependence on n so that the final expression remains comparable.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.4; method of Example 1.4.6.

Lean follows the same inequality reasoning

```
example {x y : Real} (h : x ^ 2 + y ^ 2 <= 2) :
    (x + y) ^ 2 < 5 := by
  calc
    (x + y) ^ 2 <= (x + y) ^ 2 + (x - y) ^ 2 := by extra
    _ = 2 * (x ^ 2 + y ^ 2) := by ring
    _ <= 2 * 2 := by rel [h]
    _ < 5 := by numbers
```

`extra`: elementary order facts such as adding a square.

`rel [h]`: transport a known comparison through compatible operations.

`rw` is for equality substitution; a hypothesis with $\leq$ needs order reasoning.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 1.4.7–1.4.8.

Practice 1.4: retain useful terms

1. Let $x \in \mathbb{R}$, with $x \geq 4$. Prove $x^2 - 2x \geq 8$.
2. Let $a, b \in \mathbb{R}$. Prove $a^2 + b^2 \geq 2ab$.

For each inequality, identify the nonnegative factor or square that makes the key step valid.

Practice 1.4: solution discussion

1. We know $x \geq 4 > 0$, so multiplying by x preserves $\geq$:

$$x^2 - 2x \geq 4x - 2x = 2x \geq 8.$$

2. Use the square of the difference:

$$a^2 + b^2 = (a - b)^2 + 2ab \geq 2ab.$$

The second result has no sign assumptions on a or b; the nonnegativity comes from $(a - b)^2$.

1.5 Abbreviate additive rearrangements

The book permits a short justification when a deduction needs only addition or subtraction of terms on the two sides.

```
example {x : Int} (h : x + 6 = 2) : x = -4 := by
  addarith [h]

example {u v : Real} (h : u <= 9 - v) : u + v < 10 := by
  addarith [h]
```

On paper: "From $u \leq 9 - v$, we get $u + v \leq 9 < 10$."

The course convention

For deductions requiring multiplication or division, keep the relevant calculation visible.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 1.5.

Practice 1.5: which steps are merely additive?

Assume the variables are real numbers.

1. $a - 2b = 3 \implies a = 2b + 3.$
2. $4w + 1 = 9 \implies w = 2.$
3. $m \leq 7 - n \implies m + n < 8.$

Explain each deduction. For the one that requires division, write a full chain.

Practice 1.5: solution discussion

1. Add $2b$ to both sides: an additive rearrangement.

$$a = 2b + 3.$$

2. Division by 4 is essential:

$$w = \frac{(4w + 1) - 1}{4} = \frac{9 - 1}{4} = 2.$$

3. Add n to both sides and compare constants:

$$m + n \leq 7 < 8.$$

The presence of $2b$ in the first statement does not mean that the deduction multiplies both sides.

Chapter 1: a checklist for calculation proofs

1. Identify the domain, hypotheses, and target endpoints.
2. Expose known expressions through algebraic identities.
3. Give a reason for each equality or inequality.
4. Check nonzero denominators and signs of multipliers.
5. Check that the relations combine in one direction.
6. Use Lean feedback to inspect individual steps.

Next chapter

A proof can branch, build intermediate facts, and introduce witnesses. Calculations will become components of larger arguments.

Reading and source references

Heather Macbeth, *The Mechanics of Proof*, Chapter 1:

- 1.1 Proving equalities
- 1.2 Proving equalities in Lean
- 1.3 Tips and tricks
- 1.4 Proving inequalities
- 1.5 A shortcut

https://hrmacbeth.github.io/math2001/01_Proofs_by_Calculation.html

Companion code: https://github.com/hrmacbeth/math2001

These slides provide an independently written classroom presentation and new numerical practice. Use the book for additional examples and exercises.

Chapter 2
Proofs with structure

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Based on Heather Macbeth's textbook

What you will learn

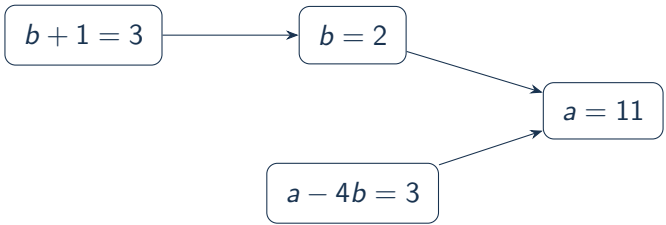
- Build and reuse an intermediate fact without losing track of the goal.
- Apply an existing lemma by checking its hypotheses.
- Use and prove statements involving $\vee$, $\wedge$, and $\exists$.
- Organize cases and witnesses in both mathematical English and Lean.

The recurring question

What information does the hypothesis provide, and what information does the goal require?

2.1 A proof can have a dependency structure

Suppose one hypothesis quickly determines b, while another relates a to b.



Prove $b = 2$ once, give it a name, and use it in the later calculation. The diagram records which facts the conclusion depends on.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.1; method of Example 2.1.1.

Worked example: an intermediate fact simplifies the calculation

Let $a, b \in \mathbb{R}$, with $a - 4b = 3$ and $b + 1 = 3$. Prove $a = 11$.

Intermediate fact

Subtracting 1 from $b + 1 = 3$ gives $b = 2$.

Now

$$\begin{aligned} a &= (a - 4b) + 4b && \text{algebra} \\ &= 3 + 4 \cdot 2 && \text{the hypothesis and intermediate fact} \\ &= 11 && \text{arithmetic.} \end{aligned}$$

The word "therefore" should identify a real dependency: this calculation uses the value of b.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.1; numerical variant of Example 2.1.1.

Lean: `have` adds a proved fact to the context

```
example {a b : Real}
    (h1 : a - 4 * b = 3) (h2 : b + 1 = 3) : a = 11 := by
  have hb : b = 2 := by addarith [h2]
  calc
    a = (a - 4 * b) + 4 * b := by ring
    _ = 3 + 4 * 2 := by rw [h1, hb]
    _ = 11 := by numbers
```

- Before `have`: only `h1` and `h2` are available.
- After it: `hb : b = 2` is also available.
- The original goal is still $a = 11$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 2.1.1; numerical variant.

Cancellation requires a reason

Example. If $t \in \mathbb{R}$, $t^2 = 4t$, and $t \geq 1$, then

$$t \cdot t = 4t, \qquad t \neq 0, \qquad \text{hence } t = 4.$$

Without $t \neq 0$, cancellation would discard the possible solution $t = 0$.

Cancelling squares is different

If $u, v \geq 0$ and $u^2 \leq v^2$, then $u \leq v$. Sign information matters: $(-3)^2 = 3^2$ does not imply $-3 = 3$.

The textbook tactic `cancel` checks the required side conditions. On paper, state them when they are doing mathematical work.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 2.1.4–2.1.5.

Practice 2.1: identify an intermediate fact

1. Let $m, n \in \mathbb{Z}$, with $m + 2 \leq 3n$ and $n \leq 4$. Prove $m \leq 10$.
2. Let $a, b \in \mathbb{R}$, with $a^2 = b^2 + 9$ and $a \geq 0$. Prove $a \geq 3$.

Write down a useful intermediate inequality before writing the final sentence. For the second problem, indicate where $a \geq 0$ is needed.

Practice 2.1: solution discussion

1. First establish $m + 2 \leq 12$:

$$m + 2 \leq 3n \leq 3 \cdot 4 = 12.$$

Subtracting 2 gives $m \leq 10$.

2. First establish the comparison of squares:

$$a^2 = b^2 + 9 \geq 9 = 3^2.$$

Since a and 3 are nonnegative, this implies $a \geq 3$. Without $a \geq 0$, the instance $a = -3, b = 0$ would contradict the conclusion.

2.2 A lemma converts one task into another

A lemma is a previously proved implication that may be reused.

Lemma	What it needs	What it gives
`ne_of_lt`	$a < b$	$a \neq b$
`ne_of_gt`	$a > b$	$a \neq b$
`le_antisymm`	$a \leq b$ and $b \leq a$	$a = b$

- `apply L` works backward: replace the goal by the hypotheses needed by L.
- `have h := L ...` works forward: store the conclusion obtained from supplied inputs.
- `exact h` closes a goal already proved by h.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.2.

Worked example: prove a stronger comparison

Let $x \in \mathbb{Q}$ and suppose $5x = 3$. To prove $x \neq 1$, it suffices to prove $x < 1$:

$$x = \frac{5x}{5} = \frac{3}{5} < 1.$$

```
example {x : Rat} (h : 5 * x = 3) : Not (x = 1) := by
  apply ne_of_lt
  calc
    x = 5 * x / 5 := by ring
    _ = 3 / 5 := by rw [h]
    _ < 1 := by numbers
```

The new goal after `apply ne_of_lt` is $x < 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.2; numerical variant of Example 2.2.1.

Worked example: squeeze a square to zero

Let $a, b \in \mathbb{R}$ and suppose $a^2 + b^2 = 0$. Prove $a^2 = 0$.

1. A square is nonnegative, so $0 \leq a^2$.
2. The other square is nonnegative, so
$$a^2 \leq a^2 + b^2 = 0.$$
3. Combining $a^2 \leq 0$ and $0 \leq a^2$ gives $a^2 = 0$.

Lean structure

`apply le_antisymm` creates two goals. Both must be proved; the first is an upper bound and the second a lower bound.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 2.2.3.

Practice 2.2: choose the lemma

1. Let $q \in \mathbb{Q}$, with $4q \leq 8$ and $2q \geq 4$. Prove $q = 2$.
2. Let $m \in \mathbb{Z}$, with $m + 3 = 7$. Prove $3m \neq 9$.

Which lemma should be applied first? What new goal or goals will it create?

Practice 2.2: solution discussion

1. Use antisymmetry. Its two required inequalities are

$$q = \frac{4q}{4} \leq \frac{8}{4} = 2, \qquad q = \frac{2q}{2} \geq \frac{4}{2} = 2.$$

Hence $q = 2$.

2. Since $m + 3 = 7$, we have $m = 4$. Therefore

$$3m = 3 \cdot 4 = 12 > 9.$$

Use `ne_of_gt` to obtain $3m \neq 9$.

2.3 “Or” has different roles in a hypothesis and a goal

$P \vee Q$ means that at least one of P, Q is true; both may be true.

Using $P \vee Q$

Prove the target assuming P. Then prove it assuming Q. Both cases are required because you do not know which holds.

Proving $P \vee Q$

It is enough to prove one alternative. Choose `left` for P or `right` for Q. The chosen alternative may depend on the current case.

Facts introduced inside one case are available only in that case.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 2.3.1–2.3.4.

Worked example: the same goal in two cases

Let $x, y \in \mathbb{R}$, and suppose $x = 2$ or $y = -2$. Prove $xy + 2x = 2y + 4$.

Case 1: $x = 2$

$$xy + 2x = 2y + 2 \cdot 2 = 2y + 4.$$

Case 2: $y = -2$

$$xy + 2x = x(-2) + 2x = 0 = 2(-2) + 4 = 2y + 4.$$

Each branch reaches the entire original goal.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.3; method of Example 2.3.1.

Worked example: create a useful disjunction

Let $x \in \mathbb{R}$ and suppose $x^2 - 5x + 6 = 0$. Prove $x = 2$ or $x = 3$.

1. Factor the known polynomial:

$$(x - 2)(x - 3) = x^2 - 5x + 6 = 0.$$

2. Use the zero-product property: $uv = 0$ implies $u = 0$ or $v = 0$.
3. If $x - 2 = 0$, then $x = 2$, establishing the left alternative.
4. If $x - 3 = 0$, then $x = 3$, establishing the right alternative.

The new disjunction is an intermediate fact; it is not an extra assumption.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.3; numerical variant of Example 2.3.4.

Lean: make the case structure visible

```
example {x : Real} (h : x ^ 2 - 5 * x + 6 = 0) :
    Or (x = 2) (x = 3) := by
  have hf : (x - 2) * (x - 3) = 0 := by
    calc
      (x - 2) * (x - 3) = x ^ 2 - 5 * x + 6 := by ring
      _ = 0 := h
  have hz := eq_zero_or_eq_zero_of_mul_eq_zero hf
  cases hz with
  | inl h2 =>
      left
      addarith [h2]
  | inr h3 =>
      right
      addarith [h3]
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.3; ASCII syntax for the case structure.

A discrete gap gives an exhaustive case split

Claim. No natural number has square 2.
For $n \in \mathbb{N}$, either $n \leq 1$ or $2 \leq n$.

- If $n \leq 1$, then $n^2 \leq 1 < 2$.
- If $2 \leq n$, then $n^2 \geq 4 > 2$.

Thus in either case $n^2 \neq 2$.

What changes for integers?

Handle negative n by considering the nonnegative number $-n$ and using $(-n)^2 = n^2$.

The split is exhaustive because no natural number lies strictly between 1 and 2.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 2.3.2 and 2.3.5.

Practice 2.3: create the right cases

1. If $x \in \mathbb{R}$ and $x^2 - 2x - 3 = 0$, prove $x = -1$ or $x = 3$.
2. For every $n \in \mathbb{N}$, prove $n^2 \neq 6$.

Explain why your cases cover all possibilities. For the first problem, say which alternative of the goal each case proves.

Practice 2.3: solution discussion

1. Since

$$(x+1)(x-3) = x^2 - 2x - 3 = 0,$$

we have $x + 1 = 0$ or $x - 3 = 0$. These give $x = -1$ and $x = 3$, respectively.

2. Split into $n \leq 2$ and $3 \leq n$.

$$n \leq 2 \implies n^2 \leq 4 < 6, \qquad 3 \leq n \implies n^2 \geq 9 > 6.$$

Every natural number lies in one of these cases.

2.4 An “and” packages two facts

$P \wedge Q$ means that both P and Q hold.

Using $P \wedge Q$

Extract both facts and use them in the same proof. For a Lean hypothesis `h`, its components are `h.1` and `h.2`.

Proving $P \wedge Q$

Prove both components. `constructor` creates two goals, or `And.intro` combines two existing proofs.

An inequality chain such as $0 < x < 1$ abbreviates $0 < x \wedge x < 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.4.

Worked example: solve for both variables

Let $x, y \in \mathbb{Q}$, with $x + y = 8$ and $x + 2y = 11$. Prove $x = 5$ and $y = 3$.

1. First find a reusable intermediate fact:

$$y = (x + 2y) - (x + y) = 11 - 8 = 3.$$

2. Use it to obtain the other component:

$$x = (x + y) - y = 8 - 3 = 5.$$

3. We have now proved both parts of the conjunction.

Establishing $y = 3$ before splitting the goal lets both parts refer to it.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.4; methods of Examples 2.4.1 and 2.4.3.

Lean: extract, derive, and repackage

```
example {x y : Rat}
    (h : And (x + y = 8) (x + 2 * y = 11)) :
    And (x = 5) (y = 3) := by
  have hy : y = 3 := by
    calc
      y = (x + 2 * y) - (x + y) := by ring
      _ = 11 - 8 := by rw [h.2, h.1]
      _ = 3 := by numbers
  have hx : x = 5 := by
    calc
      x = (x + y) - y := by ring
      _ = 8 - 3 := by rw [h.1, hy]
      _ = 5 := by numbers
  exact And.intro hx hy
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.4; new numerical example.

Practice 2.4: unpack and repack

1. Let $r, s \in \mathbb{R}$. Assume the single conjunction

$$r + s \leq 3 \ \wedge \ r - s \leq 5.$$

   Prove $r \leq 4$.

2. Let $p \in \mathbb{Z}$, with $p - 1 \geq 3$. Prove $p \geq 4 \wedge p^2 \geq 16$.

Which proof uses a conjunction? Which constructs a conjunction?

Practice 2.4: solution discussion

1. Both components of the hypothesis are available:

$$r = \frac{(r+s)+(r-s)}{2} \leq \frac{3+5}{2} = 4.$$

2. The hypothesis gives $p \geq 4$ by adding 1. In particular $p \geq 0$, so squaring preserves the comparison:

$$p^2 \geq 4^2 = 16.$$

These prove the two components of $p \geq 4 \wedge p^2 \geq 16$.

2.5 An existential statement promises a witness

$$\exists x \in D,\ P(x)$$

asserts that at least one object x in the domain D satisfies P.

Using an existential

Introduce a witness x and its promised property $P(x)$. You may use that property, but may not assume anything else about the witness.

Proving an existential

Choose a particular witness $w \in D$ and prove $P(w)$. Lean's `use w` leaves the property as the new goal.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.5.

Worked example: use a witness supplied by the hypothesis

Let $a \in \mathbb{Q}$. Suppose there is a rational number b with $a = b^2 + 2$. Prove $a > 0$.

1. Introduce such a number b, with its property $a = b^2 + 2$.
2. Every rational square is nonnegative, so

$$a = b^2 + 2 \geq 2 > 0.$$

The witness is not yours to choose

The hypothesis does not say that $b = 0$. It gives some b that satisfies the property, and the proof must work for that witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.5; numerical variant of Example 2.5.1.

Worked example: construct a point between two numbers

Let $p, q \in \mathbb{R}$ and suppose $p < q$. Prove that there exists x with $p < x < q$.

Choice of witness

Take $x = (p + q)/2$.

Check both required inequalities:

$$p = \frac{p+p}{2} < \frac{p+q}{2}, \qquad \frac{p+q}{2} < \frac{q+q}{2} = q.$$

Both steps use $p < q$ and division by the positive number 2. The domain matters: there is no integer strictly between 2 and 3.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 2.5.7.

Lean: a witness becomes a concrete verification task

```
example : Exists (fun n : Int => 9 * n = 54) := by
  use 6
  numbers

example (x : Real) : Exists (fun y : Real => y > x) := by
  use x + 1
  extra
```

In the first proof, `use 6` replaces an existence goal by $9 \cdot 6 = 54$.
In the second proof, the witness is a formula depending on the given x.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 2.5; methods of Examples 2.5.3–2.5.4.

Practice 2.5: propose and verify witnesses

1. Find integers m, n with $m^2 + n^2 = 65$.
2. Given any integer a, prove that some integers m, n satisfy

$$m^2 - n^2 = 2a + 1.$$

3. Given a real number t, prove that some real u satisfies $u < t$.

Write down the proposed witness before checking the property.

Practice 2.5: solution discussion

1. Take $m = 4$ and $n = 7$:

$$4^2 + 7^2 = 16 + 49 = 65.$$

2. Take $m = a + 1$ and $n = a$:

$$(a+1)^2 - a^2 = a^2 + 2a + 1 - a^2 = 2a + 1.$$

3. Take $u = t - 1$. Since $-1 < 0$, we have $t - 1 < t$.

Each proof contains two ingredients: an object of the required type and a verification of its property.

Source: Heather Macbeth, *The Mechanics of Proof*, Problem 2 follows Example 2.5.6.

Chapter 2: match the logical form to the proof action

Form	Use as a hypothesis	Prove as a goal
$P \vee Q$	Handle both cases	Establish one alternative
$P \wedge Q$	Extract both facts	Establish both components
$\exists x,\ P(x)$	Introduce witness and property	Choose and verify a witness

- `have` records intermediate work.
- A lemma application must discharge all of its required hypotheses.
- Case assumptions and witness properties have a scope; respect it.

Reading and source references

Heather Macbeth, *The Mechanics of Proof*, Chapter 2:

- 2.1 Intermediate steps
- 2.2 Invoking lemmas
- 2.3 “Or” and proof by cases
- 2.4 “And”
- 2.5 Existence proofs

https://hrmacbeth.github.io/math2001/02_Proofs_with_Structure.html

Companion code: https://github.com/hrmacbeth/math2001

These slides independently organize the chapter’s proof methods and include new numerical practice.

Chapter 3
Parity and divisibility

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you will learn

- Unfold evenness, oddness, divisibility, and congruence into witness statements.
- Construct new witnesses by algebraic calculation.
- Prove and use rules for modular arithmetic.
- Use residue cases to cover all integers.
- Explain how a Bézout identity justifies certain cancellation steps.

The recurring proof pattern

Unpack the hypothesis $\longrightarrow$ calculate $\longrightarrow$ identify a witness for the conclusion.

3.1 Definitions tell us what must be proved

For an integer n:

$$\begin{aligned} n \text{ is even} &\iff \exists k \in \mathbb{Z},\ n = 2k, \\ n \text{ is odd} &\iff \exists k \in \mathbb{Z},\ n = 2k + 1. \end{aligned}$$

- To prove evenness or oddness, produce an integer k.
- To use evenness or oddness, introduce the promised k and its equality.
- Different existential statements may have different witnesses, even if both are written with the letter k.

Lean: `dsimp [Int.Odd]` displays the textbook's definition.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.1.

Worked examples: zero and negative integers count too

Claim	Witness	Verification
0 is even	$k = 0$	$0 = 2 \cdot 0$
-8 is even	$k = -4$	$-8 = 2(-4)$
9 is odd	$k = 4$	$9 = 2 \cdot 4 + 1$
-7 is odd	$k = -4$	$-7 = 2(-4) + 1$

Why the domain matters

The witnesses for -8 and -7 are integers, not natural numbers. Parity is not restricted to positive integers.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.1; methods of Examples 3.1.1–3.1.2.

Worked example: turn one oddness witness into another

If $n = 2k + 1$, then

$$5n - 2 = 5(2k + 1) - 2 = 2(5k + 1) + 1.$$

Thus $5n - 2$ is odd, with witness $5k + 1$.

```
example {n : Int} (h : Int.Odd n) : Int.Odd (5 * n - 2) := by
  dsimp [Int.Odd] at *
  cases h with
  | intro k hk =>
      use 5 * k + 1
      calc
        5 * n - 2 = 5 * (2 * k + 1) - 2 := by rw [hk]
        _ = 2 * (5 * k + 1) + 1 := by ring
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.1; method of Example 3.1.3.

Worked example: every integer has one of two parities

Every integer is even or odd. Use this fact to prove $n^2 + n$ is even.

Case 1: $n = 2k$

$$n^2 + n = (2k)^2 + 2k = 2(2k^2 + k).$$

Case 2: $n = 2k + 1$

$$n^2 + n = (2k + 1)^2 + (2k + 1) = 2(2k^2 + 3k + 1).$$

Both witnesses are integers, and the cases exhaust all integers. Lean supplies the case split as `Int.even_or_odd n`.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 3.1.9; parity dichotomy revisited in Chapter 4.

Practice 3.1: name the output witness

1. If n is even, prove that $n^2 + 3$ is odd.
2. If m, n are odd integers, prove that mn is odd.

Use different letters for witnesses from independent hypotheses. End each proof by explicitly identifying an integer k in the form $2k + 1$.

Practice 3.1: solution discussion

1. Write $n = 2r$ for some $r \in \mathbb{Z}$. Then

$$n^2 + 3 = 4r^2 + 3 = 2(2r^2 + 1) + 1.$$

The output witness is $2r^2 + 1$.

2. Write $m = 2a + 1$ and $n = 2b + 1$, with $a, b \in \mathbb{Z}$. Then

$$mn = (2a + 1)(2b + 1) = 2(2ab + a + b) + 1.$$

The output witness is $2ab + a + b$.

Source: Heather Macbeth, *The Mechanics of Proof*, Problem 2 is a standard parity fact also practiced in Section 3.1.

3.2 Divisibility is an existence statement

For $a, b \in \mathbb{Z}$, define

$$a \mid b \quad \iff \quad \exists k \in \mathbb{Z},\ b = ak.$$

We say that a divides b, that a is a factor of b, or that b is a multiple of a.

- $6 \mid 42$, because $42 = 6 \cdot 7$.
- $-3 \mid 12$, because $12 = (-3)(-4)$.
- $a \mid 0$ for every a, because $0 = a \cdot 0$.
- $0 \mid b$ forces $b = 0$.

For divisibility in $\mathbb{N}$, the witness must lie in $\mathbb{N}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.2.1–3.2.2 and Section 3.2 exercises.

Worked example: retain a common factor

If $a \mid b$, write $b = ak$. Then

$$b^2 + 2b = (ak)^2 + 2ak = a(k(ak + 2)).$$

This proves $a \mid b^2 + 2b$.

```
example {a b : Int} (h : Dvd.dvd a b) :
    Dvd.dvd a (b ^ 2 + 2 * b) := by
  cases h with
  | intro k hk =>
      use k * (a * k + 2)
      calc
        b ^ 2 + 2 * b = (a * k) ^ 2 + 2 * (a * k) := by rw [hk]
        _ = a * (k * (a * k + 2)) := by ring
```

Source: Heather Macbeth, *The Mechanics of Proof*, Example 3.2.3; ASCII form of the divisibility notation.

Two basic facts and the hypotheses they need

Transitivity. For integers, $a \mid b$ and $b \mid c$ imply $a \mid c$. Indeed, if $b = ar$ and $c = bs$, then

$$c = (ar)s = a(rs).$$

A size bound in $\mathbb{N}$

If $a, b \in \mathbb{N}$, $b > 0$, and $a \mid b$, then $0 < a \leq b$.

Write $b = ak$. Since $ak > 0$, both a and k are positive. Thus $k \geq 1$, and $a = a \cdot 1 \leq ak = b$.

The condition $b > 0$ is essential: $8 \mid 0$, but $8 \nleq 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.2; witness composition and Examples 3.2.7–3.2.8.

Worked example: prove non-divisibility by a gap

Consecutive multiples test

Let $d > 0$ and $q, t \in \mathbb{Z}$. If $dq < t < d(q+1)$, then $d \nmid t$.

Why? If $t = dk$ for an integer k, dividing by positive d gives $q < k < q+1$, impossible for an integer.

Example. To show $4 \nmid -6$, observe

$$4(-2) = -8 < -6 < -4 = 4(-2+1).$$

The same test works for negative t; choose the two surrounding multiples.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.2; method of Example 3.2.6.

Practice 3.2: compose the witnesses

1. For $a, b \in \mathbb{Z}$, prove that $a \mid b$ implies $a \mid 3b - b^2$.
2. For $m, n, p \in \mathbb{Z}$, prove that $m \mid n$ and $n^3 \mid p$ imply $m^3 \mid p$.
3. Show that $5 \nmid 17$ by placing 17 between consecutive multiples.

Do not divide by a or m: those numbers may be zero. Use the equalities supplied by divisibility instead.

Practice 3.2: solution discussion

1. If $b = ak$, then

$$3b - b^2 = 3ak - a^2k^2 = a(3k - ak^2).$$

2. Write $n = mr$ and $p = n^3s$. Then

$$p = (mr)^3 s = m^3(r^3 s).$$

3. Since $5 \cdot 3 = 15 < 17 < 20 = 5(3 + 1)$, the number 17 is not a multiple of 5.

3.3 Congruence compares a difference to the modulus

For integers a, b, m, define

$$a \equiv b \pmod{m} \quad \iff \quad m \mid (a - b).$$

Equivalently, $a - b = mk$ for some $k \in \mathbb{Z}$.

Worked example

$-8 \equiv 2 \pmod{5}$, because $-8 - 2 = 5(-2)$.

- Congruent numbers need not be equal.
- For $m > 0$, congruence means having the same remainder modulo m.
- The definition also makes sense for $m = 0$: it reduces to $a = b$.

Lean's textbook notation is represented by `Int.ModEq m a b`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.3 definition and Examples 3.3.1–3.3.2.

Addition and subtraction preserve congruence

Assume $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Choose integers r, s with $a - b = mr$ and $c - d = ms$. Then

$$\begin{aligned}(a + c) - (b + d) &= (a - b) + (c - d) \\ &= mr + ms = m(r + s),\end{aligned}$$

$$\begin{aligned}(a - c) - (b - d) &= (a - b) - (c - d) \\ &= mr - ms = m(r - s).\end{aligned}$$

Therefore $a + c \equiv b + d \pmod{m}$ and $a - c \equiv b - d \pmod{m}$. Negation is similar: $(-a) - (-b) = -(a - b) = m(-r)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.3.3–3.3.5.

Multiplication preserves congruence

Again assume $a - b = mr$ and $c - d = ms$. The useful identity is

$$ac - bd = (a - b)c + b(c - d).$$

It gives

$$\begin{aligned} ac - bd &= (mr)c + b(ms) \\ &= m(rc + bs). \end{aligned}$$

Hence $ac \equiv bd \pmod{m}$, with witness $rc + bs$.

A useful special case

$a^2 - b^2 = (a - b)(a + b) = m(r(a + b))$, so $a^2 \equiv b^2 \pmod{m}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.3.6 and 3.3.8.

Congruence supports chains and powers

For integers a, b, c, m:

- **Reflexivity:** $a - a = m \cdot 0$, so $a \equiv a \pmod{m}$.
- **Symmetry:** if $a - b = mr$, then $b - a = m(-r)$.
- **Transitivity:** if $a - b = mr$ and $b - c = ms$, then $a - c = m(r + s)$.

Power rule

If $a \equiv b \pmod{m}$ and $k \in \mathbb{N}$, then $a^k \equiv b^k \pmod{m}$.

For $k = 0$ both sides are 1. Repeated multiplication proves each positive-power case; Chapter 6 organizes this argument as induction on k.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 3.3.8–3.3.10, Section 3.3 exercises; Example 6.1.3.

Ordinary division is not a congruence rule

Modulo 6 we have

$$2 \cdot 1 \equiv 2 \cdot 4 \pmod 6,$$

because $2 - 8 = -6$ is divisible by 6. But cancelling the 2 would assert

$$1 \equiv 4 \pmod 6,$$

which is false: $1 - 4 = -3$ is not divisible by 6.

What failed?

An equality can be divided by a nonzero real factor. A congruence needs additional arithmetic information about the factor and modulus.

Section 3.5 supplies identities that make certain cancellations valid.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.3; new counterexample illustrating Example 3.3.7.

Practice 3.3: prove the rule from the definition

Suppose $a \equiv b \pmod{m}$.

1. Prove $2a + 1 \equiv 2b + 1 \pmod{m}$ by giving a witness.
2. Prove $a^3 \equiv b^3 \pmod{m}$ using a factorization.

Start from $a - b = mr$. Your calculation should end with m times an integer.

Practice 3.3: solution discussion

Choose $r \in \mathbb{Z}$ with $a - b = mr$. **1.**

$$(2a + 1) - (2b + 1) = 2(a - b) = 2mr = m(2r).$$

2.

$$\begin{aligned} a^3 - b^3 &= (a - b)(a^2 + ab + b^2) \\ &= m(r(a^2 + ab + b^2)). \end{aligned}$$

The witnesses are $2r$ and $r(a^2 + ab + b^2)$, respectively.

Source: Heather Macbeth, *The Mechanics of Proof*, Problem 2 follows Example 3.3.9.

3.4 Calculate with the rules you have proved

Let $a, b \in \mathbb{Z}$, with $a \equiv 3 \pmod 7$ and $b \equiv 4 \pmod 7$. Find the residue of $a^2 + b$.

$$\begin{aligned} a^2 + b &\equiv 3^2 + 4 \pmod 7 && \text{power and addition rules} \\ &= 13 && \text{arithmetic} \\ &\equiv 6 \pmod 7 && 13 - 6 = 7 \cdot 1. \end{aligned}$$

Why this is a proof

Each congruence uses an established rule; an equality also gives a congruence. Transitivity connects the initial expression to its final residue.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.4; method of Example 3.4.2.

Worked example: cover every possible residue

Claim. Every integer x satisfies $x^3 \equiv x \pmod 3$.
There are three possible residues of x modulo 3:

Residue r	r^3	$r^3 - r$	Divisible by 3?
0	0	0	$0 = 3 \cdot 0$
1	1	0	$0 = 3 \cdot 0$
2	8	6	$6 = 3 \cdot 2$

In each case $x^3 \equiv r^3 \equiv r \equiv x \pmod 3$. The table is exhaustive because every integer has one of these residues.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 3.4.4.

Lean: congruence transport and numerical verification

```
-- In the textbook's modular-arithmetic environment:
example {a b : Int}
    (ha : Int.ModEq 7 a 3) (hb : Int.ModEq 7 b 4) :
    Int.ModEq 7 (a ^ 2 + b) (3 ^ 2 + 4) := by
  rel [ha, hb]

example : Int.ModEq 7 13 6 := by
  use 1
  numbers
```

The chapter file also imports `Library.Tactic.ModEq`.
For residue cases, `mod_cases hx : x % 3` creates the three cases used on the preceding slide.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.4 and its companion Lean file.

Practice 3.4: substitution, witnesses, and cases

1. If $n \equiv 2 \pmod 5$, show $n^2 + 3n + 1 \equiv 1 \pmod 5$.
2. Find an integer k such that $4k \equiv 3 \pmod 7$.
3. Show that every integer square is congruent to 0 or 1 modulo 4.

Use a different method in each problem: substitute, choose a witness, and split into residues.

Practice 3.4: solution discussion

1. Substitute the known residue:

$$n^2 + 3n + 1 \equiv 2^2 + 3 \cdot 2 + 1 = 11 \equiv 1 \pmod 5.$$

2. Take $k = 6$, since $4 \cdot 6 - 3 = 21 = 7 \cdot 3$.

3. For residues $r = 0, 1, 2, 3$ modulo 4:

$$r^2 = 0, 1, 4, 9 \quad \text{has residue} \quad 0, 1, 0, 1.$$

The power rule transfers this calculation from r back to any integer having residue r.

3.5 Expressing 1 as an integer combination

The integers 3 and 7 satisfy

$$(-2) \cdot 3 + 1 \cdot 7 = 1.$$

Multiplying by any integer n gives

$$n = -2(3n) + 7n.$$

Why this helps

If $3n$ is a multiple of 7, then both terms on the right are multiples of 7. Their sum n must therefore also be a multiple of 7.

A relation expressing 1 as an integer combination of two numbers is a Bézout identity for those numbers.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.5; method of Example 3.5.1.

Worked example: justify a cancellation by an explicit witness

Assume $n \in \mathbb{Z}$ and $7 \mid 3n$. Choose $t \in \mathbb{Z}$ with $3n = 7t$. Then

$$\begin{aligned} n &= -2(3n) + 7n && (-2) \cdot 3 + 7 = 1 \\ &= -2(7t) + 7n && \text{the divisibility witness} \\ &= 7(n - 2t) && \text{algebra.} \end{aligned}$$

Hence $7 \mid n$, with witness $n - 2t$.

Compare with the failed cancellation modulo 6

Here the conclusion follows from a specific integer identity. Nonzero factors alone do not justify cancellation in congruences.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.5; numerical variant of Example 3.5.2.

The general cancellation statement behind the example

Theorem

Let $m, n, u, v, k \in \mathbb{Z}$, and suppose $um + vn = 1$. If $m \mid nk$, then $m \mid k$.

Proof. Choose $t \in \mathbb{Z}$ with $nk = mt$. Then

$$\begin{aligned} k &= (um + vn)k \\ &= m(uk) + v(nk) \\ &= m(uk) + v(mt) \\ &= m(uk + vt). \end{aligned}$$

The conclusion has witness $uk + vt$.

Source: Heather Macbeth, *The Mechanics of Proof*, Generalization of the witness method in Section 3.5.

Worked example: combine two divisibility assumptions

Suppose $5 \mid z$ and $7 \mid z$. Write $z = 5a$ and $z = 7b$. Since $3 \cdot 5 - 2 \cdot 7 = 1$,

$$\begin{aligned} z &= 15z - 14z \\ &= 15(7b) - 14(5a) \\ &= 35(3b - 2a). \end{aligned}$$

Thus $35 \mid z$.

Check the arithmetic relationship first

From $4 \mid 12$ and $6 \mid 12$, we cannot conclude $24 \mid 12$. The product conclusion needs more than two divisibility assumptions.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 3.5; numerical variant of Example 3.5.3.

Practice 3.5: find an identity equal to 1

1. Prove that $6 \mid 5n$ implies $6 \mid n$, for every integer n.
2. If $4 \mid z$ and $9 \mid z$, prove that $36 \mid z$.

Useful identities to investigate:

$$6 - 5 = 1, \qquad 9 - 2 \cdot 4 = 1.$$

State the new divisibility witness explicitly.

Practice 3.5: solution discussion

1. Choose $t \in \mathbb{Z}$ with $5n = 6t$. Then

$$n = 6n - 5n = 6n - 6t = 6(n - t).$$

The new witness is $n - t$.

2. Write $z = 4a$ and $z = 9b$. Then

$$z = 9z - 8z = 9(4a) - 8(9b) = 36(a - 2b).$$

The new witness is $a - 2b$.

Chapter 3: definitions turn number theory into algebra

Statement	What you must exhibit
n is even	$n = 2k$, with $k \in \mathbb{Z}$
n is odd	$n = 2k + 1$, with $k \in \mathbb{Z}$
$a \mid b$	$b = ak$, with $k \in \mathbb{Z}$
$a \equiv b \pmod{m}$	$a - b = mk$, with $k \in \mathbb{Z}$

- Use existing witnesses to build the witness required by the conclusion.
- Congruence respects addition, subtraction, multiplication, and natural powers.
- Cancellation needs a justified arithmetic condition; a Bézout identity supplies one.

Reading and source references

Heather Macbeth, *The Mechanics of Proof*, Chapter 3:

- 3.1 Definitions; parity
- 3.2 Divisibility
- 3.3 Modular arithmetic: theory
- 3.4 Modular arithmetic: calculations
- 3.5 Bézout's identity

https://hrmacbeth.github.io/math2001/03_Parity_and_Divisibility.html

Companion code: https://github.com/hrmacbeth/math2001

These slides independently present the definitions, proofs, and new numerical exercises. The power-rule induction is developed in Chapter 6.

Chapter 4
Proofs with structure, II

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you should be able to do

- Read nested quantifiers in the order in which they bind variables.
- Prove both directions of an equivalence.
- Separate the existence and uniqueness obligations in a claim.
- Eliminate an impossible case and prove a negative statement.
- Match each logical step to a small Lean tactic.

Working convention

Domains matter: $\mathbb{N} = \{0, 1, 2, \ldots\}$; $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ have different possible witnesses. Lean snippets use the textbook's educational tactic environment.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 4, Sections 4.1–4.5.

4.1: A universal statement is a reusable rule

Statement	To prove it	To use it
$\forall x \in A,\ P(x)$	Fix an arbitrary $x \in A$.	Substitute a particular $t \in A$.
$P \to Q$	Assume P; prove Q.	Supply a proof of P to obtain Q.

$$\forall x \in \mathbb{R}, \quad x^2 \leq 2 \longrightarrow x \leq a$$

This rule only supplies $t \leq a$ after we establish $t^2 \leq 2$.

Lean connection

`intro x hx` introduces the object and its assumption. `exact h t ht` specializes a rule; `apply h` works backward.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1.

A best lower bound: choosing versus fixing a variable

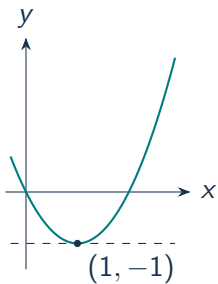
If $a \leq x^2 - 2x$ for every real x, use $x = 1$:

$$a \leq 1^2 - 2 \cdot 1 = -1.$$

Conversely, -1 is a lower bound because for an arbitrary $x \in \mathbb{R}$,

$$x^2 - 2x = (x-1)^2 - 1 \geq -1.$$

Thus the largest lower bound is -1.



Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Examples 4.1.1 and 4.1.5.

Quantifiers determine the order of the proof

Goal: $\exists b \in \mathbb{R},\ \forall x \in \mathbb{R},\ b \leq x^2 - 2x$.

```
-- Demonstration in the textbook's tactic environment.
example : Exists (fun b : Real =>
    forall x : Real, b <= x ^ 2 - 2 * x) := by
  use -1
  intro x
  calc
    -1 <= -1 + (x - 1) ^ 2 := by extra
    _ = x ^ 2 - 2 * x := by ring
```

The witness b is chosen *before* x is introduced. It cannot depend on x.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Example 4.1.5; ASCII teaching adaptation.

A quantified hypothesis can prove uniqueness

Suppose $a, b \in \mathbb{R}$ satisfy

$$a^2 \leq 2, \quad b^2 \leq 2, \qquad \forall y \in \mathbb{R},\ y^2 \leq 2 \Rightarrow y \leq a,$$

$$\forall y \in \mathbb{R},\ y^2 \leq 2 \Rightarrow y \leq b.$$

1. Apply the rule for b to $y = a$, using $a^2 \leq 2$: obtain $a \leq b$.
2. Apply the rule for a to $y = b$, using $b^2 \leq 2$: obtain $b \leq a$.
3. Antisymmetry of $\leq$ gives $a = b$.

Lean connection

`apply le_antisymm` produces the two inequalities. Each one is an application of a quantified implication.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Example 4.1.4.

“For all sufficiently large” has a quantifier order

For integers, this means

$$\exists N \in \mathbb{Z},\ \forall n \in \mathbb{Z}, \quad N \leq n \Rightarrow P(n).$$

For example, take $N = 5$ to show $n^3 \geq 4n^2 + 7$ eventually. For any integer $n \geq 5$,

$$n^3 - 4n^2 = n^2(n-4) \geq n^2 \geq 25 \geq 7.$$

The first inequality uses $n - 4 \geq 1$ and $n^2 \geq 0$.

What the proof supplies

A single threshold works for every later input. It need not be the smallest possible threshold.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Example 4.1.7.

4.1 Practice: identify the next object or assumption

1. Let $a \in \mathbb{Q}$ and suppose $a \geq -3 + 4b - b^2$ for every $b \in \mathbb{Q}$. Prove $a \geq 1$.
2. Prove $\exists n \in \mathbb{N},\ \forall m \in \mathbb{N},\ n \leq m$.
3. Prove $\exists a \in \mathbb{R},\ \forall b \in \mathbb{R},\ \exists c \in \mathbb{R},\ a + b < c$.

For each proof, state which witnesses may depend on previously introduced variables.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Exercises 4.1.10.

4.1 Solution ideas

1. Specialize the hypothesis at $b = 2$:

$$a \geq -3 + 4 \cdot 2 - 2^2 = 1.$$

2. Choose $n = 0$. Every natural number m satisfies $0 \leq m$.
3. Choose $a = 0$. After fixing an arbitrary b, choose $c = b + 1$. Then $a + b = b < b + 1 = c$.

Dependency check

In the third proof, a is independent of b; c is allowed to depend on b. Reversing quantifiers can change the problem.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.1, Exercises 4.1.10.

4.2: An equivalence has two obligations

$$P \leftrightarrow Q \quad \text{means} \quad (P \to Q) \wedge (Q \to P).$$

For $a \in \mathbb{Q}$,

$$3a + 1 \leq 7 \quad \Longleftrightarrow \quad a \leq 2.$$

Forward: subtract 1 and divide by positive 3. Backward: multiply by positive 3 and add 1.

```
-- Demonstration: each branch has its own assumption.
example (P Q : Prop) (forward : P -> Q)
    (backward : Q -> P) : Iff P Q := by
  constructor
  . exact forward
  . exact backward
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Example 4.2.1; adapted tactic demonstration.

Divisibility equivalence: the difficult direction needs an idea

For $n \in \mathbb{Z}$, prove $8 \mid 5n \iff 8 \mid n$.

Forward direction

Write $5n = 8a$ with $a \in \mathbb{Z}$. The identity $-3 \cdot 5 + 2 \cdot 8 = 1$ gives

$$n = -3(5n) + 16n = 8(-3a + 2n).$$

The integer $-3a + 2n$ is the divisibility witness.

Backward direction

If $n = 8a$, then $5n = 8(5a)$.

Do not cancel a factor in a divisibility statement without a reason; the coprimality of 5 and 8 is doing the work.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Example 4.2.2.

Solving an equation is an equivalence proof

For a real number x,

$$x^2 + x - 6 = 0 \quad \Longleftrightarrow \quad x = -3 \ \vee \ x = 2.$$

1. If the equation holds, then $(x + 3)(x - 2) = 0$.
2. The zero-product rule gives $x + 3 = 0$ or $x - 2 = 0$.
3. These equations give the two candidate roots.
4. Conversely, substitution of either -3 or 2 makes the original polynomial zero.

Lean connection

`rw [mul_eq_zero] at h` turns a zero-product hypothesis into a disjunction. Use `obtain h1 | h2 := h` to handle its branches.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Examples 4.2.5 and 4.2.7.

4.2 Practice: prove the converse too

1. For $x \in \mathbb{R}$, prove $2x - 1 = 11 \iff x = 6$.
2. For $k \in \mathbb{N}$, prove

$$k^2 \leq 6 \quad \iff \quad k = 0 \ \vee \ k = 1 \ \vee \ k = 2.$$

3. Explain why item 2 is false if $k \in \mathbb{R}$.

Label the forward and backward directions before calculating.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Exercises 4.2.10; domain check is an extension.

4.2 Solution ideas

1. Forward: $2x = 12$, so $x = 6$. Backward: $2 \cdot 6 - 1 = 11$.
2. If $k^2 \leq 6$, then $k \geq 3$ would imply $k^2 \geq 9$, a contradiction. Thus $0 \leq k \leq 2$, giving the three integer cases. Conversely, their squares are $0, 1, 4$, all at most 6.
3. Over $\mathbb{R}$, $k = \frac{1}{2}$ is a counterexample to the forward implication.

Proof habit

A finite list follows from bounds only in a discrete domain. For an equivalence, checking the listed values proves only one direction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.2, Exercises 4.2.10.

4.3: Existence plus uniqueness

$$\exists! x \in A,\ P(x) \quad \iff \quad \exists x \in A,\ P(x) \wedge \forall y \in A, (P(y) \to y = x).$$

To prove $\exists! a \in \mathbb{R},\ 3a + 1 = 7$:

1. **Choose:** $a = 2$.
2. **Verify existence:** $3 \cdot 2 + 1 = 7$.
3. **Verify uniqueness:** for any y with $3y + 1 = 7$,

$$y = \frac{(3y + 1) - 1}{3} = \frac{7 - 1}{3} = 2.$$

`use 2`, then `constructor`, separates the two obligations in Lean.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Example 4.3.1.

Use uniqueness by building a second candidate

Let $x \in \mathbb{Q}$. Suppose exactly one rational a satisfies $a^2 = x$.

1. Existence supplies a rational a with $a^2 = x$.
2. The number $-a$ is another candidate: $(-a)^2 = a^2 = x$.
3. Uniqueness forces $-a = a$.
4. Therefore $2a = 0$, so $a = 0$ and $x = a^2 = 0$.

Why existence matters

A rational number such as -1 has no rational square root. "At most one" would hold for it, but the conclusion $x = 0$ would fail.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Example 4.3.3.

A unique representative modulo 5

Prove $\exists! r \in \mathbb{Z}$ such that $0 \leq r < 5$ and $14 \equiv r \pmod 5$.

Existence

Take $r = 4$: $14 - 4 = 5 \cdot 2$ and $0 \leq 4 < 5$.

Uniqueness

If $0 \leq r < 5$ and $14 - r = 5q$, then

$$10 \leq 14 - r \leq 14, \qquad 2 \leq q < 3.$$

Since q is an integer, $q = 2$; hence $r = 14 - 10 = 4$.

Without the interval restriction, $4, 9, -1, \ldots$ all represent the same congruence class.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Example 4.3.4.

4.3 Practice: do not stop after finding a witness

1. Prove $\exists! x \in \mathbb{Q},\ 4x - 3 = 9$.
2. Prove $\exists! n \in \mathbb{N},\ \forall a \in \mathbb{N},\ n \leq a$.

For each item, write two separate sentences:

- The chosen object has the required property because ...
- Every other object with the property equals it because ...

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Exercises 4.3.5.

4.3 Solution ideas

1. Choose $x = 3$ and check $4 \cdot 3 - 3 = 9$. If $4y - 3 = 9$, then $4y = 12$ and $y = 3$.
2. Choose $n = 0$ and use $0 \leq a$ for each $a \in \mathbb{N}$. If $m \leq a$ for every natural a, specialize at $a = 0$. Then $m \leq 0$, and since $0 \leq m$, we obtain $m = 0$.

Reusable pattern

A universal property can identify a unique object by testing it at a strategically chosen input.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.3, Exercises 4.3.5.

4.4: An impossible case imposes no further obligation

If the assumptions give both P and $\neg P$, they cannot all hold. In logic, from this contradiction we may derive any target Q.

```
-- Demonstration of the logical rule.
example (P Q : Prop) (hP : P) (hnP : Not P) : Q := by
  contradiction
```

For example, $t \in \mathbb{Z}$, $t < 3$, and $t - 1 = 6$ imply both $t = 7$ and $7 < 3$.

Interpretation

The conditional theorem is valid because no input satisfies its assumptions. This is useful when eliminating a branch of a case split.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Examples 4.4.1 and 4.4.2.

Rule out the wrong sign

Let $x, y \in \mathbb{R}$, $x \geq 0$, and $xy > 0$. Prove $y > 0$.

1. Split into $y \leq 0$ or $y > 0$.
2. In the first case, multiplying by $x \geq 0$ gives

$$xy \leq x \cdot 0 = 0,$$

   contradicting $xy > 0$.
3. In the second case, the desired conclusion is already available.

Where the hypothesis enters

The condition $x \geq 0$ justifies the direction of the inequality. Without it, $x = y = -1$ would be a counterexample.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Example 4.4.1.

Residue cases can expose a contradiction

For $n \in \mathbb{Z}$, suppose $n^2 + n + 1 \equiv 1 \pmod 3$.

$n \pmod 3$	$n^2 + n + 1 \pmod 3$	Consequence
0	1	Allowed
1	0	Contradicts the assumption
2	1	Allowed

Thus $n \equiv 0 \pmod 3$ or $n \equiv 2 \pmod 3$.

Lean connection

`mod_cases` creates the residue cases. The excluded case yields $0 \equiv 1 \pmod 3$; the textbook uses `numbers at H` on that numeric contradiction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Example 4.4.3.

4.4 Practice: eliminate impossible branches

1. Let $x \in \mathbb{Q}$, $x^2 = 4$, and $x > 1$. Prove $x = 2$ by factoring and considering cases.
2. Let $n \in \mathbb{Z}$ and $n^2 \equiv 4 \pmod 5$. Prove

$$n \equiv 2 \pmod 5 \quad \vee \quad n \equiv 3 \pmod 5.$$

For each excluded case, state the contradictory pair of facts explicitly.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Exercises 4.4.6.

4.4 Solution ideas

1. $(x+2)(x-2)=0$, so $x=-2$ or $x=2$. The first case gives $1<-2$, contradicting the numerical order. Therefore $x=2$.
2. Square the five possible residues:

$n \pmod 5$	0	1	2	3	4
$n^2 \pmod 5$	0	1	4	4	1

   Only residues 2 and 3 match the given square residue 4.

A case split is complete only when every branch is either solved or contradicted.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.4, Exercises 4.4.6.

4.5: To prove a negation, assume its target

The logical meaning of $\neg P$ is $P \to \mathrm{False}$. To prove $\neg \forall x \in \mathbb{R},\ x^2 \geq x$, assume that universal claim and test $x = \frac{1}{2}$.

$$\left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4} < \tfrac{1}{2}.$$

```
-- Demonstration in the textbook's tactic environment.
example : Not (forall x : Real, x ^ 2 >= x) := by
  intro h
  have H := h (1 / 2)
  numbers at H
```

The newly introduced assumption is discharged by the proof of the negation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.1.

Why a number between consecutive multiples is not divisible

For integers a, b, q, assume $bq < a < b(q+1)$. Then $b \nmid a$.

1. The interval is nonempty, so $bq < b(q+1)$ and hence $b > 0$.
2. If $a = bk$ for some integer k, then

$$bq < bk < b(q+1) \quad \Rightarrow \quad q < k < q+1.$$

3. There is no integer strictly between consecutive integers.

Example: $3 \cdot 4 < 13 < 3 \cdot 5$, so $3 \nmid 13$.

Lean connection

`intro h` assumes divisibility; `obtain` extracts its integer witness. The contradiction is an impossible bound on that witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Examples 4.5.2 and 4.5.8.

A contradiction-based test for primality

Definition

A natural number p is prime when $p \geq 2$ and every natural divisor of p equals 1 or p.

If $p = kl$ for $k, l \in \mathbb{N}$ with $k \neq 1$ and $k \neq p$, then p is not prime: $k \mid p$, but neither allowed alternative holds. Equivalently, to prove p prime, it suffices to show

$$p \geq 2, \qquad \forall m \in \mathbb{N},\ 1 < m < p \Rightarrow m \nmid p.$$

A divisor of positive p lies between 1 and p. Eliminating the interior leaves exactly the two endpoints.

Source: Heather Macbeth, *The Mechanics of Proof*, Sections 4.1, 4.4, 4.5; Examples 4.1.8, 4.4.4, 4.5.7.

A smaller search interval for a prime test

Let $p, T \in \mathbb{N}$. Assume

$$p \geq 2, \qquad p < T^2, \qquad \forall m \in \mathbb{N},\ 1 < m < T \Rightarrow m \nmid p.$$

Then p is prime.

1. If p were composite, write $p = ml$ with $1 < m < p$.
2. Since $ml = p > m$, we have $l > 1$.
3. If both $m \geq T$ and $l \geq T$, then $p = ml \geq T^2$, a contradiction.
4. One factor lies strictly between 1 and T, contrary to the assumption.

For $p = 79$, use $T = 9$ and check divisors $2, 3, \ldots, 8$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.9.

4.5 Practice: plan the contradiction before calculating

1. Prove that no natural number n satisfies $n^2 = 2$.
2. Use the improved prime test with $T = 10$ to prove 97 prime. For each $m = 2, \ldots, 9$, locate 97 between consecutive multiples of m.

For item 1, split the natural numbers at the boundary between 1 and 2.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.4 and Exercises 4.5.10.

4.5 Solution ideas

1. If $n \leq 1$, then $n^2 \leq 1 < 2$. If $n \geq 2$, then $n^2 \geq 4 > 2$. Both cases contradict $n^2 = 2$.
2. We have $2 \leq 97 < 10^2$. The lower quotients are:

m	2	3	4	5	6	7	8	9
q	48	32	24	19	16	13	12	10
mq	96	96	96	95	96	91	96	90

In every column, $mq < 97 < m(q+1)$, so $m \nmid 97$. The improved prime test now applies.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 4.5, Example 4.5.4 and Exercises 4.5.10.

Review: let the outermost logical symbol guide you

Goal	Mathematical action	Lean cue
$\forall x,\ P(x)$	Fix an arbitrary x	`intro x`
$P \to Q$	Assume P	`intro hP`
$P \leftrightarrow Q$	Prove two implications	`constructor`
$\exists! x,\ P(x)$	Witness, property, uniqueness	`use`, `constructor`
$\neg P$	Assume P and derive falsehood	`intro hP`
Any goal with $P, \neg P$	Eliminate the impossible case	`contradiction`

Before leaving a proof, check domains, witness dependencies, both directions, and all cases.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 4, Sections 4.1–4.5.

Sources and further work

Heather Macbeth, *The Mechanics of Proof*, Chapter 4.

- 4.1: universal quantification, implication, and eventual properties.
- 4.2: equivalence and complete solution sets.
- 4.3: existence and uniqueness.
- 4.4–4.5: contradictions, nonexistence, and prime tests.

Read Chapter 4 and its remaining exercises.

These slides use original teaching prose and drawings. Example and exercise numbers in source comments identify the adaptation. Lean demonstrations use the textbook's educational tactics.

Chapter 5
Logic

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you should be able to do

- Check a propositional formula with a truth table and prove a logical equivalence.
- Track how quantifier order controls the dependence of witnesses.
- Use excluded middle without first deciding which case is true.
- Move negations through connectives and quantifiers correctly.
- Turn a negative statement into a concrete proof task.

Three levels of reasoning

Truth tables analyze propositions. Quantifiers introduce objects. Mathematics supplies facts about those objects.

Lean fragments use the textbook's educational tactic environment.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 5, Sections 5.1–5.3.

5.1: The connectives and their truth conditions

Let P and Q be propositions.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \to Q$	$P \leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

The one false implication case

$P \to Q$ fails exactly when P is true and Q is false. When P is false, there is no instance of P that violates the implication.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.2 and Exercise 5.1.3.

Build a compound truth table from the inside out

Consider $P \leftrightarrow (\neg P \vee Q)$.

P	Q	$\neg P$	$\neg P \vee Q$	$P \leftrightarrow (\neg P \vee Q)$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	F
F	F	T	T	F

Compute $\neg P$ first, then the disjunction, then the equivalence. The final column matches $P \wedge Q$.

Check versus explanation

For propositional formulas, matching truth tables establishes classical equivalence. A structural proof explains how to convert a proof in each direction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Exercise 5.1.3 and Example 5.1.5.

Logical equivalence means mutual implication

Two statements A and B are logically equivalent when $A \leftrightarrow B$. For example, $(P \vee P) \leftrightarrow P$.

```
-- Demonstration: every branch of the disjunction gives P.
example (P : Prop) : Iff (Or P P) P := by
  constructor
  . intro h
    obtain h1 | h2 := h
    . exact h1
    . exact h2
  . intro h
    left
    exact h
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.4; ASCII teaching adaptation.

Distributing a conjunction over a disjunction

$$P \wedge (Q \vee R) \quad \Longleftrightarrow \quad (P \wedge Q) \vee (P \wedge R).$$

Forward

We have P and either Q or R.

- If Q, build $P \wedge Q$ and choose the left alternative.
- If R, build $P \wedge R$ and choose the right alternative.

Backward

Split the given disjunction.

- From $P \wedge Q$, keep P and introduce $Q \vee R$ on the left.
- From $P \wedge R$, keep P and introduce $Q \vee R$ on the right.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.5.

A common witness works for every input

Let X, Y be types and let $P(x, y)$ be a proposition for $x \in X$, $y \in Y$.

$$(\exists x \in X,\ \forall y \in Y,\ P(x, y)) \quad \Longrightarrow \quad (\forall y \in Y,\ \exists x \in X,\ P(x, y)).$$

1. Obtain a witness x_0 that works for every y.
2. Fix an arbitrary y.
3. Use that same x_0; the universal hypothesis gives $P(x_0, y)$.

The converse can fail. Over $\mathbb{Z}$, take $P(x, y)$ to mean $x > y$. For each y, $x = y + 1$ works; no one integer is greater than every integer.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.6; converse counterexample is an extension.

Negating existence: two direct implications

$$\neg \exists x \in A,\ P(x) \quad \Longleftrightarrow \quad \forall x \in A,\ \neg P(x).$$

Forward

Assume no witness exists. For an arbitrary $a \in A$, if $P(a)$ held, a would be a witness. Thus $\neg P(a)$.

Backward

Assume every $a \in A$ fails P. If a witness w existed, we would have both $P(w)$ and $\neg P(w)$.

This equivalence also holds when A is empty. No choice of a default element is needed.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.6.

5.1 Practice: build both conversions

1. Prove $(P \wedge P) \leftrightarrow P$.
2. Prove $\neg(P \vee Q) \leftrightarrow (\neg P \wedge \neg Q)$.
3. Suppose $\forall x \in A,\ P(x) \leftrightarrow Q(x)$. Prove

$$(\exists x \in A,\ P(x)) \leftrightarrow (\exists x \in A,\ Q(x)).$$

In item 3, must the witness change when you convert the statement?

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Exercises 5.1.7.

5.1 Solution ideas

1. Forward: keep one copy of P. Backward: use the proof of P twice to build $P \wedge P$.
2. Forward: a proof of either P or Q would give the forbidden disjunction. Backward: split a hypothetical $P \vee Q$; each branch contradicts the corresponding negation.
3. Forward: obtain w and $P(w)$, apply the forward implication at w, and keep w as the witness for Q. Backward: use the reverse implication at the same witness.

The common structure is to unpack the given statement and rebuild the target.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Exercises 5.1.7.

5.2: A case split need not decide the proposition

Law of the excluded middle

For every proposition P, we may reason by the two cases P and $\neg P$.

To prove a target Q:

1. Assume P and prove Q.
2. Assume $\neg P$ and prove Q.
3. Conclude Q, because the cases cover all possibilities.

In Lean, `by_cases h : P` creates these two branches. This is a classical logical principle, not an algorithm for deciding arbitrary propositions.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Examples 5.2.5 and 5.2.6.

A predicate with easy endpoints and a difficult middle

Following the textbook, call $k \in \mathbb{N}$ *superpowered* when

$$S(k): \quad \forall n \in \mathbb{N}, \quad k^{k^n} + 1 \text{ is prime.}$$

Exponentiation is grouped as $k^{(k^n)}$; the convention is $k^0 = 1$.

k	Useful calculation	Conclusion
0	$0^{0^0} + 1 = 1$	$\neg S(0)$
1	$1^{1^n} + 1 = 2$ for every n	$S(1)$
3	$3^{3^0} + 1 = 4$	$\neg S(3)$

For $k = 2$, the first values are $3, 5, 17, 257, 65537$. A finite list of successes does not prove $S(2)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Examples 5.2.1–5.2.4.

Prove a boundary exists without locating it first

We know $S(1)$ and $\neg S(3)$. Prove

$$\exists k \in \mathbb{N}, \quad S(k) \wedge \neg S(k+1).$$

Case	Choose	Why it works
$S(2)$	$k = 2$	$S(2)$ and $\neg S(3)$
$\neg S(2)$	$k = 1$	$S(1)$ and $\neg S(2)$

Both cases produce a witness. This argument proves existence even before we determine which case holds.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Example 5.2.5.

The same boundary argument in abstract Lean

```
-- Demonstration: the argument uses only S 1 and Not (S 3).
example (S : Nat -> Prop) (h1 : S 1) (h3 : Not (S 3)) :
    Exists (fun k => And (S k) (Not (S (k + 1)))) := by
  by_cases h2 : S 2
  . use 2
    constructor
    . exact h2
    . exact h3
  . use 1
    constructor
    . exact h1
    . exact h2
```

The logical structure does not depend on the arithmetic inside S.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Example 5.2.5; abstract teaching adaptation.

Double negation and contraposition

Double negation elimination

Given $\neg\neg P$, split into P or $\neg P$. The first case proves P directly. The second contradicts $\neg\neg P$. Thus P follows classically.

Contraposition

$$(P \to Q) \quad \iff \quad (\neg Q \to \neg P).$$

Forward: assuming $\neg Q$ and P, the implication supplies the forbidden Q. Backward: assume P. If Q failed, the contrapositive would give $\neg P$; hence Q holds by classical reasoning.

The forward implication is direct; the reverse uses double negation elimination.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Example 5.2.6 and Exercises 5.2.7.

5.2 Practice: identify the classical step

1. From $\neg P \to \neg Q$ and Q, prove P. Show explicitly where excluded middle is used.
2. You are given the exact factorization

$$2^{2^5} + 1 = 4294967297 = 641 \cdot 6700417.$$

   Use it to select a concrete k satisfying $S(k) \wedge \neg S(k+1)$, where S is the superpowered predicate.

Why are five prime values insufficient to establish a universal statement?

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Exercises 5.2.7.

5.2 Solution ideas

1. Split into P and $\neg P$. In the first case, we are done. In the second, the given implication produces $\neg Q$, contradicting Q. Excluded middle licenses that case split.
2. The value at $n = 5$ has a nontrivial factor 641, so $\neg S(2)$. We already know $S(1)$; therefore choose $k = 1$.

What a counterexample must provide

To refute $\forall n,\ P(n)$, one failing input is enough. To prove it, checking a fixed finite sample is generally insufficient.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Exercises 5.2.7.

5.3: Push negations through propositional operations

Original	Equivalent form	Textbook Lean lemma
$\neg\neg P$	P	`not_not`
$\neg(P \vee Q)$	$\neg P \wedge \neg Q$	`not_or`
$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	`not_and_or`
$\neg(P \to Q)$	$P \wedge \neg Q$	`not_imp`

Interpret the last row

An implication fails when its premise holds and its conclusion fails. Negating an implication does *not* negate both sides.

These are classical equivalences; some directions require excluded middle.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Table 5.1.

Negating quantifiers changes who chooses the object

$$\neg \exists x \in A,\ P(x) \quad \Longleftrightarrow \quad \forall x \in A,\ \neg P(x),$$
$$\neg \forall x \in A,\ P(x) \quad \Longleftrightarrow \quad \exists x \in A,\ \neg P(x).$$

- No witness works $\Rightarrow$ every candidate fails.
- Not every candidate works $\Rightarrow$ there is a counterexample.

For ordered numbers, also use

$$\neg(a < b) \Longleftrightarrow b \leq a, \qquad \neg(a \leq b) \Longleftrightarrow b < a.$$

The domain of each quantifier stays the same throughout the transformation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Table 5.1 and Example 5.3.3.

Normalize one connective at a time

Start with

$$\neg(\forall m \in \mathbb{Z},\ m \neq 2 \Rightarrow \exists n \in \mathbb{Z},\ n^2 = m).$$

1. Negate the universal quantifier:

$$\exists m \in \mathbb{Z},\ \neg(m \neq 2 \Rightarrow \exists n \in \mathbb{Z},\ n^2 = m).$$

2. Negate the implication:

$$\exists m \in \mathbb{Z},\ m \neq 2 \wedge \neg\exists n \in \mathbb{Z},\ n^2 = m.$$

3. Negate the existential quantifier:

$$\exists m \in \mathbb{Z},\ m \neq 2 \wedge \forall n \in \mathbb{Z},\ n^2 \neq m.$$

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.2.

Why negating “and” requires a case split

Prove $\neg(P \wedge Q) \Rightarrow \neg P \vee \neg Q$.

1. Suppose $\neg(P \wedge Q)$.
2. By excluded middle, either P or $\neg P$.
3. If $\neg P$, choose the left alternative.
4. If P, show $\neg Q$: an assumed Q would combine with P to give the forbidden $P \wedge Q$. Choose the right alternative.

Conversely, if either P or Q is false, they cannot both hold.

Lean connection

Use `constructor`, then `by_cases hP : P` in the forward direction; use `obtain` in the reverse direction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.1.

A chained inequality hides a conjunction

The statement $n^2 < m < (n+1)^2$ means

$$n^2 < m \ \wedge \ m < (n+1)^2.$$

Therefore

$$\neg(\forall n \in \mathbb{Z},\ \exists m \in \mathbb{Z},\ n^2 < m < (n+1)^2)$$

is equivalent to

$$\exists n \in \mathbb{Z},\ \forall m \in \mathbb{Z}, \quad m \leq n^2 \ \vee \ (n+1)^2 \leq m.$$

To *prove* the resulting statement, choose $n = 0$. Every integer m satisfies $m \leq 0$ or $1 \leq m$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.3; witness proof is an extension.

Normalize first, then supply the mathematical idea

```
-- Demonstration in the textbook's tactic environment.
example : Not (forall x : Real, x ^ 2 >= x) := by
  push_neg
  use (1 / 2 : Real)
  numbers
```

After `push_neg`, the goal asks for a real x with $x^2 < x$. The tactic exposes the witness obligation; the choice $x = \frac{1}{2}$ is the mathematical step.

What normalization does

It changes the form of the statement by equivalences. It does not automatically find the counterexample or prove the resulting inequality.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.4 and Exercises 5.3.6.

5.3 Practice: normalize, then prove or refute

1. Move all negations inward:

$$\neg\big(\forall a, b \in \mathbb{Z},\ ab = 1 \Rightarrow (a = 1 \vee b = 1)\big).$$

   Then decide whether the statement is true.

2. Normalize

$$\neg\big(\exists x \in \mathbb{R},\ \forall y \in \mathbb{R},\ y \leq x\big),$$

   and prove the normalized statement.

Keep the variable domains visible while changing the logical symbols.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Exercises 5.3.6.

5.3 Solution ideas

1. The normalized form is

$$\exists a, b \in \mathbb{Z}, \quad ab = 1 \wedge a \neq 1 \wedge b \neq 1.$$

   It is true: choose $a = b = -1$. Their product is 1, and neither equals 1.

2. The normalized form is

$$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R}, \quad y > x.$$

   For an arbitrary x, choose $y = x + 1$. Then $y > x$.

A successful normalization turns the original wording into explicit inputs and witnesses.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Exercises 5.3.6.

One theorem, two useful proof presentations

$$\neg\exists n \in \mathbb{N},\ n^2 = 2 \quad \Longleftrightarrow \quad \forall n \in \mathbb{N},\ n^2 \neq 2.$$

Proof of the normalized form

Fix $n \in \mathbb{N}$. Either $n \leq 1$ or $n \geq 2$.

$$n \leq 1 \Rightarrow n^2 \leq 1 < 2, \qquad n \geq 2 \Rightarrow n^2 \geq 4 > 2.$$

Either strict inequality implies $n^2 \neq 2$.

The numerical argument is unchanged from Chapter 4, but normalization lets us begin with an arbitrary input rather than a hypothetical witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.5.

Review: four questions before manipulating a statement

1. What is the outermost connective or quantifier?
2. Am I proving an implication, an equivalence, or the truth of a particular formula?
3. Which variables may a witness depend on?
4. Does this direction use excluded middle?

A reliable workflow

Write the domains $\longrightarrow$ expose hidden conjunctions $\longrightarrow$ normalize one layer at a time $\longrightarrow$ identify the resulting proof obligations.

A truth table checks propositional structure; a quantified claim also needs reasoning about its objects.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 5, Sections 5.1–5.3.

Sources and further work

Heather Macbeth, *The Mechanics of Proof*, Chapter 5.

- 5.1: logical equivalence, truth tables, and quantifiers.
- 5.2: excluded middle, the superpowered predicate, and contraposition.
- 5.3: Table 5.1 and worked negation normalizations.

Read Chapter 5 and practice the remaining logical equivalences.

Teaching prose, tables, and added counterexamples are original. Source comments record example locations. Lean snippets use the textbook's educational tactics.

Chapter 6
Induction

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

6.1 Induction: an infinite chain of implications

For a proposition $Q(n)$ indexed by $n \in \mathbb{N} = \{0, 1, 2, \ldots\}$, prove

$$Q(0), \qquad \forall k \in \mathbb{N}, \quad Q(k) \Longrightarrow Q(k+1).$$

These two facts establish $Q(n)$ for every natural number n.

Worked example: $2^n \geq n + 1$ for every $n \in \mathbb{N}$

Base: $2^0 = 1 = 0 + 1$. Fix $k \in \mathbb{N}$ and assume $2^k \geq k + 1$. Then

$$2^{k+1} = 2 \cdot 2^k \geq 2(k+1) = (k+2) + k \geq k + 2.$$

The last inequality uses $k \geq 0$; the first uses the hypothesis at k.

The induction hypothesis is available at k, while the goal concerns $k + 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Example 6.1.1.

Lean: identify the base and successor goals

Here is a closely related claim, $n + 1 \leq 3^n$, over $\mathbb{N}$.

```
example (n : Nat) : n + 1 <= 3 ^ n := by
  simple_induction n with k IH
  . numbers
  . calc
      k + 1 + 1 <= 3 * (k + 1) := by addarith
      _ <= 3 * 3 ^ k := by rel [IH]
      _ = 3 ^ (k + 1) := by ring
```

- `simple_induction` gives the base goal and a successor goal with `IH`.
- `rel [IH]` transports an inequality through multiplication by 3.
- `addarith` handles linear arithmetic; `ring` handles the identity.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Example 6.1.1.

Powers preserve congruence: prove the general rule

Let $a, b, m \in \mathbb{Z}$ with $a \equiv b \pmod{m}$. We prove $a^n \equiv b^n \pmod{m}$ for every $n \in \mathbb{N}$.
Choose $v \in \mathbb{Z}$ with $a - b = mv$.
Base: $a^0 - b^0 = 1 - 1 = m \cdot 0$.
Step: fix $k \in \mathbb{N}$ and assume $a^k - b^k = mu$ for some $u \in \mathbb{Z}$. Then

$$\begin{aligned} a^{k+1} - b^{k+1} &= a(a^k - b^k) + b^k(a - b) \\ &= a(mu) + b^k(mv) \\ &= m(au + b^k v). \end{aligned}$$

The new witness $au + b^k v$ is an integer. Induction completes the proof. No division by m is used, so the argument also covers $m = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Example 6.1.3; general rule used in Chapter 3.

Starting at a threshold

"For all sufficiently large n" means

$$\exists N \in \mathbb{N}\, \forall n \in \mathbb{N}, \quad N \leq n \Longrightarrow Q(n).$$

For $Q(n) : 2^n \geq n^2$, choose $N = 4$. The base is $16 = 4^2$. For $k \geq 4$, assume $2^k \geq k^2$. Since

$$2k^2 - (k+1)^2 = k(k-2) - 1 \geq 4 \cdot 2 - 1 = 7,$$

we get $2^{k+1} \geq 2k^2 \geq (k+1)^2$.

Lean setup pattern for a goal with `hn : 4 <= n`:

```
induction_from_starting_point n, hn with k hk IH
```

The step now includes `hk : 4 <= k`; the base is at 4, not 0.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1, Examples 6.1.5–6.1.6.

Practice 6.1: a disjunctive invariant

Prove, for every $n \in \mathbb{N}$,

$$5^n \equiv 1 \pmod 8 \quad \text{or} \quad 5^n \equiv 5 \pmod 8.$$

All powers and congruences here are interpreted in $\mathbb{Z}$.

1. State the base case at $n = 0$.
2. Write the exact induction hypothesis at k.
3. Split that hypothesis into two cases. Which alternative follows at $k + 1$?

In Lean, a disjunctive hypothesis can be split using `obtain h1 | h5 := IH`; choose the goal's alternative using `left` or `right`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1.7.

Solution 6.1: multiplication swaps the alternatives

Base: $5^0 = 1$, so choose the first alternative.
Fix $k \in \mathbb{N}$ and assume $5^k \equiv 1$ or $5^k \equiv 5 \pmod 8$.

- If $5^k \equiv 1$, then
$$5^{k+1} = 5 \cdot 5^k \equiv 5 \cdot 1 = 5 \pmod 8.$$
- If $5^k \equiv 5$, then
$$5^{k+1} = 5 \cdot 5^k \equiv 25 \equiv 1 \pmod 8,$$
since $25 - 1 = 8 \cdot 3$.

Thus the same disjunction holds at $k + 1$. Induction completes the proof.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.1.7.

6.2 Recursion defines; induction proves

A sequence with values in a set S is a function $\mathbb{N} \to S$. A recurrence specifies initial values and rules for later values.
For example, define $b : \mathbb{N} \to \mathbb{Z}$ by

$$b_0 = 3, \qquad b_{n+1} = b_n^2 - 2 \quad (n \in \mathbb{N}).$$

Its first terms are $3, 7, 47, 2207$. Each call uses $n < n + 1$.

Invariant: every b_n is odd

Base: $b_0 = 2 \cdot 1 + 1$. For $k \in \mathbb{N}$, assume $b_k = 2t + 1$ for some $t \in \mathbb{Z}$. Then

$$b_{k+1} = (2t + 1)^2 - 2 = 2(2t^2 + 2t - 1) + 1.$$

The displayed integer $2t^2 + 2t - 1$ is the new oddness witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Example 6.2.1.

A closed formula from a shifted recurrence

Let $x_0 = 5$ and $x_{n+1} = 2x_n - 1$, with $x : \mathbb{N} \to \mathbb{Z}$. Subtract the fixed point 1: $x_{n+1} - 1 = 2(x_n - 1)$.

```
def x : Nat -> Int
  | 0 => 5
  | k + 1 => 2 * x k - 1

example (n : Nat) : x n - 1 = 4 * 2 ^ n := by
  simple_induction n with k IH
  . rw [x]; numbers
  . calc
      x (k + 1) - 1 = 2 * (x k - 1) := by rw [x]; ring
      _ = 2 * (4 * 2 ^ k) := by rw [IH]
      _ = 4 * 2 ^ (k + 1) := by ring
```

Hence $x_n = 2^{n+2} + 1$, and $x_n \equiv 1 \pmod 4$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Examples 6.2.2–6.2.3.

A recursively defined sum

Define $A : \mathbb{N} \to \mathbb{Q}$ by $A_0 = 0$ and $A_{n+1} = A_n + (n+1)$ for $n \in \mathbb{N}$. Thus $A_n = 1 + 2 + \cdots + n$, with the empty sum equal to 0.

Claim: $A_n = \dfrac{n(n+1)}{2}$ for all $n \in \mathbb{N}$

Base: $A_0 = 0 = 0(0+1)/2$. For $k \in \mathbb{N}$, assume $A_k = k(k+1)/2$. Then

$$\begin{aligned} A_{k+1} &= A_k + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Lean connection: `rw [A]` exposes the recurrence, `rw [IH]` replaces the smaller sum, and `ring` finishes in $\mathbb{Q}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Example 6.2.4.

Factorials: keep the divisor universally quantified

Define $0! = 1$ and $(n+1)! = (n+1)n!$ for $n \in \mathbb{N}$. Induction also gives $n! \geq 1$: the base is 1, and both factors in $(k+1)k!$ are at least 1.

Claim: $\forall n, d \in \mathbb{N},\ 1 \leq d \leq n \implies d \mid n!$

Induct on n, keeping the claim about *every* such d. At $n = 0$, no d satisfies $1 \leq d \leq 0$. At $n = k+1$, let $1 \leq d \leq k+1$.

- If $d = k+1$, then $(k+1)! = d\,k!$.
- Otherwise $d \leq k$. The IH gives $k! = dt$ for some $t \in \mathbb{N}$, so $(k+1)! = d((k+1)t)$.

In Lean, introduce d *after* `simple_induction n`.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2, Example 6.2.5.

Practice 6.2: formulas and bounds

1. Define $c : \mathbb{N} \to \mathbb{Z}$ by

$$c_0 = 7, \qquad c_{n+1} = 3c_n - 10 \quad (n \in \mathbb{N}).$$

   Prove $c_n = 2 \cdot 3^n + 5$ for all $n \in \mathbb{N}$. What fixed point should you subtract from the recurrence?

2. Prove $(n+1)! \geq 2^n$ for every $n \in \mathbb{N}$. Write the successor factorial so that the induction hypothesis applies.

For each proof, name the base case and state the IH before calculating.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2.7; Example 6.2.6.

Solution 6.2: expose one recursive layer

1. Closed formula. The fixed point is 5, since $3 \cdot 5 - 10 = 5$. Base: $7 = 2 \cdot 3^0 + 5$. For $k \in \mathbb{N}$, assuming $c_k = 2 \cdot 3^k + 5$ gives

$$c_{k+1} = 3(2 \cdot 3^k + 5) - 10 = 2 \cdot 3^{k+1} + 5.$$

2. Factorial bound. Base: $1! = 1 = 2^0$. For $k \in \mathbb{N}$, assuming $(k+1)! \geq 2^k$ gives

$$(k+2)! = (k+2)(k+1)! \geq (k+2)2^k \geq 2 \cdot 2^k = 2^{k+1}.$$

The second inequality uses $k + 2 \geq 2$ and $2^k \geq 0$.
Both inductions follow the decreasing index in the recursive definition.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.2.7; Example 6.2.6.

6.3 Two base cases, two induction hypotheses

To establish $Q(n)$ for all $n \in \mathbb{N}$, it suffices to prove

$$Q(0), \quad Q(1), \quad \forall k \in \mathbb{N},\ (Q(k) \wedge Q(k+1)) \Longrightarrow Q(k+2).$$

Let $a : \mathbb{N} \to \mathbb{Z}$ satisfy $a_0 = 2$, $a_1 = 1$, $a_{n+2} = a_{n+1} + 2a_n$. Claim: $a_n = 2^n + (-1)^n$.
Bases: $2 = 1 + 1$ and $1 = 2 - 1$. **Step:** for $k \in \mathbb{N}$, assume the formula at both k and $k+1$.

$$\begin{aligned} a_{k+2} &= (2^{k+1} + (-1)^{k+1}) + 2(2^k + (-1)^k) \\ &= 4 \cdot 2^k + (-1)^k = 2^{k+2} + (-1)^{k+2}. \end{aligned}$$

Both recursive arguments, k and $k+1$, are smaller than $k+2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.1.

Fibonacci: an exponential upper bound

Define $F : \mathbb{N} \to \mathbb{Z}$ by

$$F_0 = F_1 = 1, \qquad F_{n+2} = F_{n+1} + F_n \quad (n \in \mathbb{N}).$$

The first terms are $1, 1, 2, 3, 5, 8, 13$.

Claim: $F_n \leq 2^n$ for all $n \in \mathbb{N}$

Bases: $F_0 = 1 \leq 1$ and $F_1 = 1 \leq 2$. For $k \in \mathbb{N}$, assume $F_k \leq 2^k$ and $F_{k+1} \leq 2^{k+1}$. Then

$$F_{k+2} \leq 2^{k+1} + 2^k = 3 \cdot 2^k \leq 4 \cdot 2^k = 2^{k+2}.$$

The bound at $k+1$ alone would leave F_k uncontrolled.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.3.

Sometimes the induction hypothesis needs more information

For the sequence $a_0 = 2$, $a_1 = 1$, $a_{n+2} = a_{n+1} + 2a_n$, prove $a_n \equiv 1$ or $5 \pmod 6$ for $n \geq 1$.
Use the stronger assertion that the pair (a_n, a_{n+1}) is congruent to

$$(1,5) \quad \text{or} \quad (5,1) \pmod 6.$$

Base $n = 1$: $(a_1, a_2) = (1,5)$. For $k \geq 1$, assume this assertion at k. The recurrence gives

$$(1,5) \longmapsto (5, 5 + 2 \cdot 1) \equiv (5,1),$$

$$(5,1) \longmapsto (1, 1 + 2 \cdot 5) \equiv (1,5).$$

Ordinary induction now works on the *pair assertion*; its first component implies the original claim. This excludes the unwanted pairs $(1,1)$ and $(5,5)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.2.

Two-step induction from a later starting point

Define $d : \mathbb{N} \to \mathbb{Z}$ by $d_0 = 3$, $d_1 = 1$, $d_{n+2} = 3d_{n+1} + 5d_n$. To prove $d_n \geq 4^n$ for $n \geq 4$, check *two* bases:

$$d_4 = 267 \geq 256, \qquad d_5 = 1096 \geq 1024.$$

For $k \geq 4$, the hypotheses at $k, k+1$ give

$$d_{k+2} \geq 3 \cdot 4^{k+1} + 5 \cdot 4^k = 17 \cdot 4^k \geq 4^{k+2}.$$

Lean setup patterns:

```
two_step_induction n with k IH0 IH1
-- Or, with hn : 4 <= n:
two_step_induction_from_starting_point n, hn with k hk IH0 IH1
```

Here `IH0` concerns k and `IH1` concerns $k+1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3, Example 6.3.5.

Practice 6.3: a second-order recurrence

Define $t : \mathbb{N} \to \mathbb{Z}$ by

$$t_0 = 5, \qquad t_1 = 7, \qquad t_{n+2} = 2t_{n+1} - t_n \quad (n \in \mathbb{N}).$$

Prove $t_n = 2n + 5$ for every $n \in \mathbb{N}$.

1. Check both bases.
2. State separate hypotheses for t_k and t_{k+1}.
3. Substitute both formulas into the recurrence.
4. Explain why checking only t_0 cannot determine the whole sequence.

When expressing the formula in Lean, use an integer cast of n.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3.6.

Solution 6.3: the two initial values matter

Bases: $t_0 = 5 = 2 \cdot 0 + 5$ and $t_1 = 7 = 2 \cdot 1 + 5$.
Let $k \in \mathbb{N}$. Assume $t_k = 2k + 5$ and $t_{k+1} = 2(k+1) + 5$. Then

$$\begin{aligned} t_{k+2} &= 2(2(k+1) + 5) - (2k + 5) \\ &= 2k + 9 = 2(k+2) + 5. \end{aligned}$$

The recurrence has the same first difference at each step: $t_{n+2} - t_{n+1} = t_{n+1} - t_n$. Changing t_1 to 8 while keeping $t_0 = 5$ would instead produce $5, 8, 11, 14, \ldots$. The value at 0 does not fix the initial difference.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.3.6.

6.4 Strong induction: use any smaller input

For $Q : \mathbb{N} \to \mathrm{Prop}$, prove for every $n \in \mathbb{N}$:

$$(\forall m \in \mathbb{N},\ m < n \Longrightarrow Q(m)) \Longrightarrow Q(n).$$

Then $Q(n)$ holds for all n. At $n = 0$ there are no smaller inputs.

Worked proof: every $n \geq 2$ has a prime divisor

IH: every m with $2 \leq m < n$ has a prime divisor. If n is prime, use $p = n$, since $n = n \cdot 1$. Otherwise, the proper-factor lemma gives $n = mx$ with $2 \leq m < n$. The IH gives $m = py$ for a prime p, hence $n = p(yx)$.

The least relevant input $n = 2$ is prime. The recursive branch uses $m < n$, which need not be $m = n - 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4, Examples 6.4.1–6.4.2.

Lean: a theorem can call itself on smaller inputs

Using the preceding definition of F:

```
theorem fib_upper (n : Nat) : F n <= (2 : Int) ^ n := by
  match n with
  | 0 => rw [F]; numbers
  | 1 => rw [F]; numbers
  | k + 2 =>
      have h0 := fib_upper k
      have h1 := fib_upper (k + 1)
      calc
        F (k + 2) = F (k + 1) + F k := by rw [F]
        _ <= 2 ^ (k + 1) + 2 ^ k := by rel [h0, h1]
        _ = 3 * 2 ^ k := by ring
        _ <= 3 * 2 ^ k + 2 ^ k := by extra
        _ = 2 ^ (k + 2) := by ring
```

Both calls decrease the natural-number input. A call at $k + 2$ would be circular.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4, Example 6.4.1.

Practice 6.4: remove all factors of two

Show that every $n \in \mathbb{N}$ with $n > 0$ admits a representation

$$n = 2^a x, \qquad a, x \in \mathbb{N}, \qquad x \text{ odd}.$$

An odd natural number has the form $2u + 1$ with $u \in \mathbb{N}$.

1. What witnesses work when n is odd?
2. If $n = 2m$ is positive, prove $0 < m < n$ before invoking the IH.
3. Convert a representation of m into one for n.

The measure is the positive input n; the recursive call is at $m = n/2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4.3.

Solution 6.4: split on parity

Use strong induction. At a positive n, assume the statement for every m with $0 < m < n$.

- If n is odd, choose $a = 0$, $x = n$: $n = 2^0 n$.
- If n is even, write $n = 2m$. Positivity gives $m \geq 1$; then $m < 2m = n$. By the IH, $m = 2^a x$ for some odd x. Thus

$$n = 2m = 2 \cdot 2^a x = 2^{a+1} x.$$

  Use witnesses $a + 1$ and x.

At $n = 1$ the odd branch closes directly. At $n = 0$ the premise $n > 0$ is false, so no recursive call or witnesses are required.

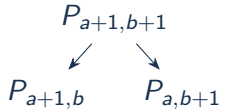
Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.4.3.

6.5 Pascal's triangle in two coordinates

Define $P : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$P(a, 0) = P(0, b) = 1, \qquad P(a+1, b+1) = P(a+1, b) + P(a, b+1).$$

```
def pascal : Nat -> Nat -> Nat
  | a, 0 => 1
  | 0, b + 1 => 1
  | a + 1, b + 1 =>
      pascal (a + 1) b + pascal a (b + 1)
termination_by _ a b => a + b
```



Each dependency has coordinate sum $a + b + 1$, less than $a + b + 2$.

For example: $P(1, 1) = 2$, $P(1, 2) = 3$, $P(2, 2) = 3 + 3 = 6$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5, Definition 6.5.1.

The factorial identity: choose the exact invariant

For all $a, b \in \mathbb{N}$,

$$\boxed{P(a,b)\, a!\, b! = (a+b)!}.$$

Induct on $a+b$, allowing the IH at *every* pair with smaller sum.

- If $b = 0$, then $P(a,0)a!0! = 1 \cdot a! \cdot 1 = (a+0)!$.
- If $a = 0$, then $P(0,b)0!b! = 1 \cdot 1 \cdot b! = (0+b)!$.

For the interior pair $(a+1, b+1)$, the two needed instances are

$$P(a+1,b)(a+1)!b! = (a+b+1)!,$$

$$P(a,b+1)a!(b+1)! = (a+b+1)!.$$

Each is justified by a decrease of one in the coordinate sum.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5, Example 6.5.3.

The factorial identity: align the two hypotheses

Write $U = P(a+1, b)$, $V = P(a, b+1)$, and $s = a + b + 1$. The IH says $U(a+1)!b! = s!$ and $Va!(b+1)! = s!$. Then

$$\begin{aligned} P(a+1,b+1)(a+1)!(b+1)! &= (U+V)(a+1)!(b+1)! \\ &= (b+1)\big[U(a+1)!b!\big] \\ &\quad + (a+1)\big[Va!(b+1)!\big] \\ &= (b+1)s! + (a+1)s! = (s+1)!. \end{aligned}$$

Since $a!b! \geq 1$, the identity implies $P(a,b) \leq (a+b)!$. Since factorials are nonzero, division in $\mathbb{Q}$ gives $P(a,b) = (a+b)!/(a!b!)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5, Examples 6.5.2–6.5.3.

Practice 6.5: an edge and a symmetry

For all natural numbers a, b, prove:

1. $P(a, 1) = a + 1$, using ordinary induction on a.
2. $P(a, b) = P(b, a)$, using induction on $a + b$.

For the symmetry proof, state the IH for arbitrary pairs (x, y) with $x + y < a + b$. Handle the two boundary directions before the interior.

Lean connection: unfold `pascal` with `rw [pascal]`. For the interior symmetry case, recursively use the symmetry theorem at $(a + 1, b)$ and $(a, b + 1)$, then use commutativity of addition.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5.4.

Solution 6.5: respect both coordinates

Edge. Base: $P(0,1) = 1$. Assume $P(k,1) = k+1$. Then

$$P(k+1,1) = P(k+1,0) + P(k,1) = 1 + (k+1) = k+2.$$

Symmetry. If either coordinate is zero, both sides are 1. For $(a+1, b+1)$, assume symmetry for all smaller sums. Then

$$\begin{aligned} P(a+1,b+1) &= P(a+1,b) + P(a,b+1) \\ &= P(b,a+1) + P(b+1,a) \\ &= P(b+1,a) + P(b,a+1) \\ &= P(b+1,a+1). \end{aligned}$$

Both IH calls have sum $a+b+1 < a+b+2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.5.4.

6.6 Recursive division on all integers

Let $M(n, d) = \operatorname{mod}(n, d)$ and $D(n, d) = \operatorname{div}(n, d)$, both functions $\mathbb{Z}^2 \to \mathbb{Z}$. Use the first applicable row:

Condition	$M(n, d)$	$D(n, d)$
$nd < 0$	$M(n + d, d)$	$D(n + d, d) - 1$
$0 < d(n - d)$	$M(n - d, d)$	$D(n - d, d) + 1$
$n = d$	0	1
otherwise	n	0

For $d > 0$, these respectively address $n < 0$, $n > d$, $n = d$, and $0 \leq n < d$.
The definitions also cover $d \leq 0$. For example, $M(n, 0) = n$, while $D(0, 0) = 1$ and $D(n, 0) = 0$ for $n \neq 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Definition 6.6.1.

Why recursive division terminates

Use the nonnegative integer measure $\mu(n, d) = |2n - d|$. The value of n itself may increase or be negative.

- If $nd < 0$, the call is at $(n + d, d)$. The difference of squared measures is

$$(2n + d)^2 - (2n - d)^2 = 8nd < 0.$$

- If $d(n - d) > 0$, the call is at $(n - d, d)$, and

$$(2n - 3d)^2 - (2n - d)^2 = -8d(n - d) < 0.$$

In both cases, the new absolute value is strictly smaller. The terminal rows make no calls. Example: dividing -7 by 3 visits $-7, -4, -1, 2$; the measures are $17, 11, 5, 1$. Thus $M(-7, 3) = 2$ and $D(-7, 3) = -3$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Definition 6.6.1.

Correctness, part I: reconstruct the dividend

Claim: $M(n,d) + dD(n,d) = n$ for all $n, d \in \mathbb{Z}$. Induct on $|2n - d|$. The IH is this identity at every (m, c) with $|2m - c| < |2n - d|$.

- If $nd < 0$, put $M' = M(n + d, d)$ and $D' = D(n + d, d)$. The IH says $M' + dD' = n + d$, so

$$M(n,d) + dD(n,d) = M' + d(D' - 1) = (n + d) - d = n.$$

- If $d(n - d) > 0$, use the IH $M' + dD' = n - d$ at $(n - d, d)$:

$$M(n,d) + dD(n,d) = M' + d(D' + 1) = (n - d) + d = n.$$

- The $n = d$ row gives $0 + d \cdot 1 = n$; the final row gives $n + d \cdot 0 = n$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Example 6.6.2.

Correctness, part II: bound the remainder

Claim: for $n, d \in \mathbb{Z}$ with $d > 0$,

$$0 \leq M(n, d) < d.$$

Induct on $|2n - d|$, with this range statement as the IH at all smaller inputs with positive divisor.

- In either recursive row, the divisor stays $d > 0$ and the remainder is unchanged. Apply the IH to $(n + d, d)$ or $(n - d, d)$.
- If $n = d$, the remainder is 0, and $0 < d$.
- In the final row, the failed tests give $nd \geq 0$ and $d(n - d) \leq 0$. Since $d > 0$, these imply $0 \leq n \leq d$. The failed equality test gives $n \neq d$, hence $n < d$. Here $M(n, d) = n$.

The positive-divisor premise is used to cancel d without reversing inequalities.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Examples 6.6.3–6.6.4.

Lean names and the existence theorem

The textbook calls M `fmod` and D `fdiv`. These are the relevant lemma applications:

```
-- Context: n d : Int, hd : 0 < d
have hid := fmod_add_fdiv n d
have hlo := fmod_nonneg_of_pos n hd
have hhi := fmod_lt_of_pos n hd
```

Together they give witnesses $q = D(n, d)$ and $r = M(n, d)$ for

$$\forall n, d \in \mathbb{Z}, \quad d > 0 \Longrightarrow \exists q, r \in \mathbb{Z}, \quad n = r + dq \ \wedge \ 0 \leq r < d.$$

Also $n - r = dq$, so $n \equiv r \pmod{d}$: every integer has a representative among $0, 1, \ldots, d - 1$.

The recursive proofs use `split_ifs`, `cancel`, and the source's `termination_by _ n d => 2 * n - d`; integer size means absolute value.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6, Examples 6.6.2–6.6.5.

Practice 6.6: uniqueness and a negative input

1. Follow the recursive definition to compute $M(-8, 3)$ and $D(-8, 3)$. Record $|2n - 3|$ at each call.
2. Let $n, d, q, q', r, s \in \mathbb{Z}$ with $d > 0$ and

$$n = r + dq = s + dq', \qquad 0 \leq r < d, \quad 0 \leq s < d.$$

   Prove $r = s$ and then $q = q'$.

Hint: subtract the reconstruction equations. How large can $r - s$ be? Which integer multiples of d fit strictly between $-d$ and d?

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6.6.

Solution 6.6: a bounded multiple must be zero

Computation: the inputs are $-8, -5, -2, 1$ with divisor 3. The measures are $19, 13, 7, 1$. At 1, the output is $(M, D) = (1, 0)$; three returns subtract 1 from D, giving

$$M(-8, 3) = 1, \qquad D(-8, 3) = -3, \qquad -8 = 1 + 3(-3).$$

Uniqueness: subtraction gives $r - s = d(q' - q)$, and the bounds imply $-d < r - s < d$. Put $t = q' - q \in \mathbb{Z}$.

- If $t \geq 1$, then $dt \geq d$, impossible.
- If $t \leq -1$, then $dt \leq -d$, impossible.

Thus $t = 0$, so $q' = q$ and $r = s$. The existence theorem plus this argument gives a unique quotient–remainder pair.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.6.6.

6.7 Euclid's algorithm, including negative inputs

For $a, b \in \mathbb{Z}$, define $G(a,b) = \gcd(a,b)$ recursively by

$$G(a,b) = \begin{cases} G(b, M(a,b)), & b > 0, \\ G(b, M(a,-b)), & b < 0, \\ a, & b = 0,\ a \geq 0, \\ -a, & b = 0,\ a < 0. \end{cases}$$

The measure is $|b|$. For $b \neq 0$, put $r = M(a,|b|)$. The remainder theorem gives $0 \leq r < |b|$, so $|r| < |b|$.

Nonnegativity proof: induct on $|b|$ with IH $G(x,y) \geq 0$ whenever $|y| < |b|$. If $b = 0$, $G(a,0) = |a| \geq 0$; otherwise $G(a,b) = G(b,r) \geq 0$ by the IH. In particular, $G(0,0) = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Definition 6.7.1; Example 6.7.2.

Why the output divides both inputs

Claim: $G(a,b) \mid a$ and $G(a,b) \mid b$ for all $a, b \in \mathbb{Z}$. Induct on $|b|$, assuming *both divisibility claims* for all smaller second-coordinate absolute values.
For $b \neq 0$, let $r = M(a,|b|)$, $q = D(a,|b|)$, $g = G(b,r)$. The IH gives $b = gu$ and $r = gv$ for some $u, v \in \mathbb{Z}$.

$$a = \begin{cases} r + bq = g(v + uq), & b > 0, \\ r - bq = g(v - uq), & b < 0. \end{cases}$$

Thus g divides a as well as b, and $g = G(a,b)$.
If $b = 0$, $G(a,0) = |a|$: use witness 1 for $a \geq 0$, witness -1 for $a < 0$, and witness 0 for divisibility of $b = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Example 6.7.3.

Extended Euclid: compute coefficients along with the gcd

Define $L, R : \mathbb{Z}^2 \to \mathbb{Z}$ mutually. For $b \neq 0$, put $r = M(a, |b|)$ and $q = D(a, |b|)$.

Condition	$L(a,b)$	$R(a,b)$
$b > 0$	$R(b,r)$	$L(b,r) - qR(b,r)$
$b < 0$	$R(b,r)$	$L(b,r) + qR(b,r)$
$b = 0,\ a \geq 0$	1	0
$b = 0,\ a < 0$	-1	0

Both functions only call the pair (b, r), whose measure $|r|$ is below $|b|$.
The intended invariant is

$$L(a,b)a + R(a,b)b = G(a,b).$$

The change of sign in R matches $a = r + bq$ versus $a = r - bq$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Definition 6.7.4.

Worked computation: descend, then recover coefficients

For $(a, b) = (-26, 18)$, divide with positive divisors:

$$-26 = 10 + 18(-2), \quad 18 = 8 + 10(1), \quad 10 = 2 + 8(1), \quad 8 = 0 + 2(4).$$

The second inputs have measures $18, 10, 8, 2, 0$, so the gcd is 2. Recover (L, R) from the last row upward:

(a, b)	q	$L(a, b)$	$R(a, b)$
$(2, 0)$	—	1	0
$(8, 2)$	4	0	1
$(10, 8)$	1	1	-1
$(18, 10)$	1	-1	2
$(-26, 18)$	-2	2	3

Check: $2(-26) + 3(18) = 2$. For example, the last coefficient is $-1 - (-2) \cdot 2 = 3$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Definition 6.7.4 (numerical variant).

Proving the coefficient identity: positive second input

Theorem: $L(a,b)a + R(a,b)b = G(a,b)$ for all $a, b \in \mathbb{Z}$. Induct on $|b|$. The IH is

$$L(x,y)x + R(x,y)y = G(x,y) \quad \text{whenever } |y| < |b|.$$

If $b > 0$, let $r = M(a,b)$ and $q = D(a,b)$, so $a = r + bq$. Write $\ell = L(b,r)$ and $t = R(b,r)$. The IH gives $\ell b + tr = G(b,r)$. Therefore

$$\begin{aligned} L(a,b)a + R(a,b)b &= ta + (\ell - qt)b \\ &= t(r + bq) + (\ell - qt)b \\ &= tr + \ell b = G(b,r) = G(a,b). \end{aligned}$$

The cancellation of tbq explains the definition of $R(a,b)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Example 6.7.5.

The negative branch and the two terminal cases

If $b < 0$, let $r = M(a, -b)$ and $q = D(a, -b)$. Now $a = r - bq$. Put $\ell = L(b, r)$ and $t = R(b, r)$. The IH still says $\ell b + tr = G(b, r)$, since $|r| < |b|$.

$$\begin{aligned} L(a,b)a + R(a,b)b &= ta + (\ell + qt)b \\ &= t(r - bq) + (\ell + qt)b \\ &= tr + \ell b = G(b, r) = G(a, b). \end{aligned}$$

For $b = 0$ there is no recursive call:

$$\begin{aligned} a \geq 0: &\quad 1 \cdot a + 0 \cdot b = a = G(a, 0), \\ a < 0: &\quad (-1) \cdot a + 0 \cdot b = -a = G(a, 0). \end{aligned}$$

These cases complete the induction for all integer signs.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Example 6.7.5.

Lean: mutual recursion and a reusable existence theorem

The source defines `L` and `R` in a `mutual` block. Its `gcd_dvd_left` and `gcd_dvd_right` proofs also use mutual induction. All recursive calls decrease $|b|$.

Bézout's identity: for every $a, b \in \mathbb{Z}$, there are $x, y \in \mathbb{Z}$ with $xa + yb = G(a, b)$. Choose $x = L(a, b)$, $y = R(a, b)$.

```
-- In the source's Euclid namespace:
example (a b : Int) :
    Exists (fun x : Int => Exists (fun y : Int =>
      x * a + y * b = gcd a b)) := by
  refine Exists.intro (L a b) ?_
  refine Exists.intro (R a b) ?_
  exact L_mul_add_R_mul a b
```

The witnesses need not be unique: replacing (x, y) by $(x + tb, y - ta)$ for $t \in \mathbb{Z}$ preserves the equation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7, Examples 6.7.3–6.7.6.

Practice 6.7: certify the gcd

1. Compute $G(14, -8)$, $L(14, -8)$, and $R(14, -8)$. Start with $14 = 6 + 8 \cdot 1$. Apply the negative-b branch at the first step. Verify the resulting linear combination.
2. Let $a, b, d \in \mathbb{Z}$. Prove

$$d \mid a \ \wedge \ d \mid b \quad \Longrightarrow \quad d \mid G(a, b).$$

   Use Bézout's identity and divisibility witnesses, without a new induction.

Explain why this property, common divisibility, and nonnegativity characterize the gcd even when a or b is negative.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7.7; Definition 6.7.4.

Solution 6.7: coefficients certify maximality

Computation. The recursive pairs are

$$(14, -8) \to (-8, 6) \to (6, 4) \to (4, 2) \to (2, 0).$$

The quotients (using $|b|$) are $1, -2, 1, 2$. Backward coefficient pairs are

$$(1, 0),\ (0, 1),\ (1, -1),\ (-1, -1),\ (-1, -2).$$

Thus $G(14, -8) = 2$ and $(-1)14 + (-2)(-8) = 2$.

Divisibility. Write $a = du$, $b = dv$ and choose Bézout witnesses x, y. Then

$$G(a, b) = xa + yb = xdu + ydv = d(xu + yv).$$

If $(a, b) \neq (0, 0)$, the gcd is positive, so every positive common divisor is at most it. At $(0, 0)$ the gcd is 0; the divisibility characterization still holds, although there is no largest positive common divisor.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 6.7.7.

References

Heather Macbeth, *The Mechanics of Proof*, Chapter 6, "Induction."

- Sections 6.1–6.3: introduction, recurrence relations, two-step induction.
- Sections 6.4–6.5: strong induction and Pascal's triangle.
- Sections 6.6–6.7: the Division Algorithm and the Euclidean algorithm, including the extended algorithm and Bézout's identity.

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Chapter 7
Number theory

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you should be able to do

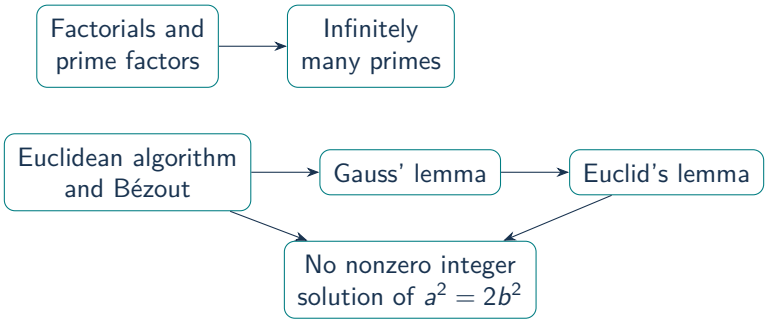
- Explain why a prime divisor of $N! + 1$ must exceed N.
- Use a Bézout identity to justify cancellation in divisibility.
- Derive Euclid's lemma from the defining property of a prime.
- Identify the decreasing positive integer in an infinite-descent proof.
- Connect an integer equation to irrationality.

Capstone perspective

The new results combine familiar tools: witnesses, case splits, induction, gcd, and contradiction. Each proof also needs a specific mathematical construction.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 7, Sections 7.1–7.3.

A map of the ingredients



For the descent proof, the needed parity fact can also be obtained directly from Chapter 6.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 7; prerequisites in Sections 6.2, 6.4, and 6.7.

7.1: What does it mean to have infinitely many primes?

Prime

A natural number p is prime if $p \geq 2$ and each natural divisor of p is either 1 or p.

We will prove the concrete statement

$$\forall N \in \mathbb{N}, \quad \exists p \in \mathbb{N}, \quad p > N \text{ and } p \text{ is prime.}$$

Why is that enough? Any finite list of natural numbers has an upper bound N. The theorem supplies a prime outside that list. The construction will produce a *prime factor* of a new number, rather than require that the new number itself be prime.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1, theorem on infinitely many primes; definition from 4.1.8.

Build a number with a prime factor

Fix an arbitrary $N \in \mathbb{N}$ and put $M = N!$.

1. $M \geq 1$, so $M + 1 \geq 2$.
2. Every natural number at least 2 has a prime factor. Choose a prime p with $p \mid M+1$.
3. Write $M + 1 = pk$ with $k \in \mathbb{N}$.
4. Since $M + 1 > 0$, $k \neq 0$. Write $k = \ell + 1$ with $\ell \in \mathbb{N}$.

We now know

$$p \geq 2, \qquad M + 1 = p(\ell + 1).$$

The next step will prove that p cannot also divide M.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; uses factorial positivity and the prime-factor theorem.

The new prime factor is larger than the bound

From $M + 1 = p(\ell + 1) = p\ell + p$ and $p \geq 2$, we get

$$p\ell < M < p(\ell + 1).$$

Thus M lies strictly between consecutive multiples of p, so $p \nmid M$.

1. Suppose $p \leq N$.
2. Since $1 \leq p \leq N$, the factorial-divisibility theorem gives $p \mid N! = M$.
3. This contradicts $p \nmid M$.

Therefore $p > N$. Since N was arbitrary, there are infinitely many primes.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; uses Examples 4.5.8 and 6.2.5.

The construction can start with a composite number

N	$N! + 1$	Prime factor p	Comparison
4	$25 = 5^2$	5	$5 > 4$
5	$121 = 11^2$	11	$11 > 5$
6	$721 = 7 \cdot 103$	7	$7 > 6$

What the theorem needs

The integer $N! + 1$ only needs to be at least 2 so that it has a prime divisor. Every such prime divisor exceeds N.

The proof gives a new prime beyond a chosen bound; it does not claim that factorial-plus-one is always prime.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; original numerical illustrations.

7.1 Practice: explain the construction's logic

1. Run the proof for $N = 0$. What are $N! + 1$ and its prime factor?
2. Suppose $p_1, \ldots, p_r$ is a nonempty finite list containing every prime. Set

$$K = p_1 p_2 \cdots p_r + 1.$$

   Explain why a prime divisor of K cannot belong to the list.

You may use: if an integer divides two integers, it divides their difference.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; classroom extensions of its construction.

7.1 Solution ideas

1. $0! = 1$, so $0! + 1 = 2$. The prime 2 is larger than 0; no special exception is needed.
2. $K \geq 2$, so choose a prime $q \mid K$. If $q = p_i$ for some listed prime, then q divides $p_1 \cdots p_r$ as well. Hence
$$q \mid (K - p_1 \cdots p_r) = 1,$$
which is impossible for $q \geq 2$.

The factorial version excludes every prime up to N; the product version excludes every prime in the given list.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.1; solutions to the preceding extensions.

7.2: When may a factor be cancelled in divisibility?

Gauss' lemma

Let $a, b, d \in \mathbb{Z}$. If

$$d \mid ab \qquad \text{and} \qquad \gcd(a, d) = 1,$$

then $d \mid b$.

The condition $\gcd(a, d) = 1$ says that a and d are coprime.

Why the hypothesis matters

$6 \mid 2 \cdot 3$, but $6 \nmid 3$. Here $\gcd(2, 6) = 2$, so the coprimality assumption fails.

Bézout's identity will express 1 as a combination of a and d.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Gauss' lemma; counterexample is an extension.

Gauss' lemma: construct the divisibility witness

Choose $x, y \in \mathbb{Z}$ with $xa + yd = 1$, using Bézout and $\gcd(a, d) = 1$. Choose $z \in \mathbb{Z}$ with $ab = dz$, using $d \mid ab$. Then

$$\begin{aligned} b &= b(xa + yd) \\ &= x(ab) + byd \\ &= x(dz) + byd \\ &= d(xz + by). \end{aligned}$$

Thus $xz + by \in \mathbb{Z}$ witnesses $d \mid b$.

Lean connection

`obtain` extracts the Bézout coefficients and divisibility witness; `use` supplies the new witness; `rw` and `ring` verify the calculation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, proof of Gauss' lemma.

A concrete cancellation problem

Let $n \in \mathbb{Z}$ and suppose $35 \mid 12n$. Prove $35 \mid n$.

1. The identity $3 \cdot 12 - 35 = 1$ supplies the needed coprimality information.
2. Write $12n = 35k$ with $k \in \mathbb{Z}$.
3. Multiply the identity by n:

$$\begin{aligned} n &= 3(12n) - 35n \\ &= 3(35k) - 35n \\ &= 35(3k - n). \end{aligned}$$

The proof not only shows divisibility; it gives a quotient $3k - n$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Gauss' lemma; original worked instance.

Euclid's lemma: primes divide one factor

Theorem

For $a, b, p \in \mathbb{N}$, if p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Put $d = \gcd(a, p)$. We know $d \mid a$ and $d \mid p$. Since p is prime, $d = 1$ or $d = p$.

Case	Conclusion
$d = 1$	Gauss' lemma applied to $p \mid ab$ gives $p \mid b$.
$d = p$	Since $d \mid a$, we have $p \mid a$.

Primality reduces the gcd to exactly the two cases needed by the proof.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Euclid's lemma.

A prime divisor of a power divides the base

Let $a, p, k \in \mathbb{N}$, with p prime and $k \geq 1$. Then

$$p \mid a^k \quad \Longrightarrow \quad p \mid a.$$

Prove this by induction on k, starting at 1.

- **Base:** $a^1 = a$.
- **Step:** if $p \mid a^{t+1} = a \cdot a^t$, Euclid's lemma gives $p \mid a$ or $p \mid a^t$. The first case finishes; the second uses the inductive hypothesis.

A consequence we will use

If a^2 is even, then a is even: apply the result with the prime $p = 2$ and $k = 2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, power corollary.

Lean keeps the integer and natural-number statements separate

The textbook defines its gcd on $\mathbb{Z}$, while its prime predicate is on $\mathbb{N}$.

1. Cast a and p to integers to apply gcd and Gauss' lemmas.
2. Use $\gcd(a, p) \geq 0$ to represent that integer gcd by a natural number d.
3. Transfer the divisibility facts to $\mathbb{N}$, where the prime definition applies.

Textbook notation/tactic	Purpose
`(a : Int)`	View a natural number as an integer.
`lift`	Replace a nonnegative integer by a natural number.
`norm_cast`	Transport compatible statements across casts.

The mathematical case split remains $d = 1$ or $d = p$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2, Lean discussion following Euclid's lemma.

7.2 Practice: choose the correct divisibility tool

1. Let $n \in \mathbb{Z}$ and $9 \mid 4n$. Use $7 \cdot 4 - 3 \cdot 9 = 1$ to produce a witness for $9 \mid n$.
2. Let $u, v \in \mathbb{N}$. If $7 \mid u^2 v$ and $7 \nmid u$, prove $7 \mid v$.
3. Find a composite d and natural a, b such that $d \mid ab$, but $d \nmid a$ and $d \nmid b$.

Name every primality or coprimality hypothesis used in your proof.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2; original classroom exercises.

7.2 Solution ideas

1. Write $4n = 9k$. Then
$$n = 7(4n) - 27n = 9(7k - 3n).$$
So $7k - 3n$ is a witness.
2. Euclid's lemma gives $7 \mid u^2$ or $7 \mid v$. The first case implies $7 \mid u$ by the power corollary, contradicting the hypothesis. Hence $7 \mid v$.
3. Take $d = 6$, $a = 2$, and $b = 3$. This shows why Euclid's lemma requires a prime divisor.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.2; solutions to the preceding exercises.

7.3: The arithmetic statement behind irrationality

Theorem

There are no $a, b \in \mathbb{N}$ with $b \neq 0$ such that $a^2 = 2b^2$.

The assumption $b \neq 0$ is essential: $(a, b) = (0, 0)$ satisfies the equation. We will prove by strong induction on b the property

$$P(b): \quad \forall a \in \mathbb{N}, \quad b \neq 0 \Rightarrow a^2 \neq 2b^2.$$

Why quantify over all numerators?

The next solution will have a different numerator. The induction hypothesis must cover every a at each smaller denominator.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, natural-number theorem and descent proof.

From one hypothetical solution, build a smaller one

Suppose $a, b \in \mathbb{N}$, $b \neq 0$, and $a^2 = 2b^2$.

1. $2 \mid a^2$, so $2 \mid a$. Write $a = 2k$ for some $k \in \mathbb{N}$.
2. Substitute and cancel the nonzero factor 2:

$$2b^2 = a^2 = (2k)^2 = 4k^2 \quad \Longrightarrow \quad b^2 = 2k^2.$$

3. Thus (b, k) satisfies the same form of equation as (a, b).

The remaining obligations

To use the induction hypothesis, we must prove $k \neq 0$ and $k < b$. A new solution alone does not yet give a contradiction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, natural-number descent; parity fact also follows from 7.2.

Check the denominator really decreases

We have $b \neq 0$ and $b^2 = 2k^2$.

1. If $k = 0$, then $b^2 = 0$, hence $b = 0$, a contradiction. Thus $k > 0$.
2. Since $k^2 > 0$,
$$k^2 < 2k^2 = b^2.$$
Because $k, b \geq 0$, this implies $k < b$.
3. The strong-induction hypothesis at denominator k, with numerator b, says $b^2 \neq 2k^2$.

This contradicts the new equation. Therefore the assumed original solution cannot exist.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, well-foundedness lemma and natural-number proof.

The precise role of the induction hypothesis

At denominator b, the hypothesis covers *all* pairs (r, s) with $s < b$:

$$\forall r, s \in \mathbb{N}, \quad s < b \Rightarrow s \neq 0 \Rightarrow r^2 \neq 2s^2.$$

Once the preceding slides supply $k < b$, $k \neq 0$, and $b^2 = 2k^2$:

```
-- Local demonstration fragment in the induction step:
-- IH covers every numerator r at every denominator s < b.
-- hkb : k < b; hk : Not (k = 0)
-- hnew : b ^ 2 = 2 * k ^ 2
have forbidden := IH b k hkb hk
exact forbidden hnew
```

The textbook expresses this as a recursive theorem call, with termination measured by the second argument.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, theorem irrat_aux and its termination explanation.

A second proof: use gcd and Bézout

Now let $a, b \in \mathbb{Z}$ with $b \neq 0$, and suppose $a^2 = 2b^2$. Put $d = \gcd(a, b)$.

1. Since $d \mid a$ and $d \mid b$, choose $k, \ell \in \mathbb{Z}$ with

$$a = dk, \qquad b = d\ell.$$

2. Bézout gives $x, y \in \mathbb{Z}$ with $ax + by = d$.
3. $d \neq 0$: otherwise $b = d\ell = 0$, a contradiction.

We will construct an integer z with $z^2 = 2$, which is impossible.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, integer theorem and Bézout proof; notation normalized.

The key identity creates an integer square root of 2

Set $z = 2\ell x + ky \in \mathbb{Z}$. Then

$$\begin{aligned} d^2 z^2 &= (2bx + ay)^2 \\ &= 2(ax + by)^2 + (y^2 - 2x^2)(a^2 - 2b^2) \\ &= 2d^2 + (y^2 - 2x^2) \cdot 0 \\ &= 2d^2. \end{aligned}$$

The middle equality is a polynomial identity, verified by expansion. Since $d \neq 0$, cancellation of d^2 gives $z^2 = 2$.

Where each input was used

$a = dk$, $b = d\ell$: first line. $ax + by = d$: third line. $a^2 = 2b^2$: third line. $d \neq 0$: cancellation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, integer proof using Brahmagupta's identity.

Why the integer square equation is impossible

If $z \in \mathbb{Z}$, then $|z|$ is a nonnegative integer. Exactly one of the following ranges applies:

Range	Bound on z^2
$\lvert z\rvert \leq 1$	$z^2 = \lvert z\rvert^2 \leq 1 < 2$
$\lvert z\rvert \geq 2$	$z^2 = \lvert z\rvert^2 \geq 4 > 2$

Thus $z^2 \neq 2$ for every integer z, contradicting the constructed z.

Conclusion of the second proof

There are no integers a, b with $b \neq 0$ and $a^2 = 2b^2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, completion of the integer proof; prerequisite Example 2.3.5.

Extension: translating the equation into irrationality

A real number is *rational* when it can be written a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$. A real number that is not rational is *irrational*. Suppose a rational $q = a/b$ satisfied $q^2 = 2$.

Multiplication by b^2 would give

$$a^2 = 2b^2,$$

contradicting the integer theorem.

Consequence

Every real solution of $q^2 = 2$ is irrational. In particular, the positive solution $\sqrt{2}$ is irrational.

No assumption that a/b is in lowest terms is needed by these proofs.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3, arithmetic theorem; rational-number interpretation added for CSC302.

7.3 Practice: audit a descent argument

1. In the natural-number proof, explain why the induction hypothesis must allow a numerator different from the original a.
2. A proposed proof produces another solution (a', b') but only shows $b' \leq b$. Explain why this does not finish a descent argument.
3. Adapt the descent proof to show that no natural a, b with $b \neq 0$ satisfy $a^2 = 3b^2$. You may use Euclid's lemma for the prime 3.

For item 3, list the new pair and prove both required inequalities about its denominator.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3; original proof-audit and generalization exercises.

7.3 Solution ideas

1. The new pair is (b, k), so the numerator changes from a to b. The induction property is quantified over every numerator.
2. A process may repeat the same denominator forever. Strong induction requires a *strictly* smaller nonnegative integer.
3. From $3 \mid a^2$, obtain $a = 3k$. Then

$$a^2 = 3b^2 \Rightarrow 9k^2 = 3b^2 \Rightarrow b^2 = 3k^2.$$

Here $k \neq 0$ because $b \neq 0$, and $k^2 < 3k^2 = b^2$ gives $k < b$. The pair (b, k) contradicts the induction hypothesis at k.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 7.3; solutions to the preceding exercises.

Review: the construction and the reason it works

Result	Construction	Decisive fact
Infinitely many primes	Prime divisor of $N! + 1$	It cannot divide $N!$.
Gauss' lemma	Multiply a Bézout identity by b	An explicit multiple of d results.
Euclid's lemma	Examine $\gcd(a, p)$	A prime has only two natural divisors.
No $a^2 = 2b^2$, $b \neq 0$	Smaller solution, or an integer square root of 2	Strict descent, or the gap between 1 and 4.

In every proof, identify the witness, preserve its domain, and check the assumption that permits cancellation or induction.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 7, Sections 7.1–7.3.

Sources and further work

Heather Macbeth, *The Mechanics of Proof*, Chapter 7.

- 7.1: factorial construction for infinitely many primes.
- 7.2: Gauss' lemma, Euclid's lemma, and the power corollary.
- 7.3: the natural-number descent and integer Bézout proofs.

Read Chapter 7 and compare the mathematical and Lean proofs.

The prose, diagrams, and classroom exercises in this deck are original adaptations. The rational-number interpretation is an explicit extension. Lean fragments use the textbook's educational environment.

Chapter 8
Functions

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What we will learn

- Read a function's *domain*, *codomain*, and rule together.
- Prove or refute injectivity, surjectivity, and bijectivity.
- Use composition and inverses to organize proofs.
- Work with ordered pairs and a bijection $\mathbb{N}^2 \to \mathbb{N}$.

A function includes its types

$f : X \to Y$ assigns exactly one element $f(x) \in Y$ to every $x \in X$. Its *image* is the set of outputs actually attained; its codomain is all of Y.

The rules $n \mapsto 2n$ on $\mathbb{Z}$ and $x \mapsto 2x$ on $\mathbb{R}$ have different surjectivity behavior.

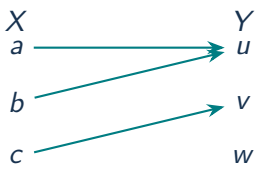
Source: Heather Macbeth, *The Mechanics of Proof*, Section 8.1.

8.1: Injectivity asks whether outputs identify inputs

Definition

$f : X \to Y$ is *injective* when

$$\forall x_1, x_2 \in X, \qquad f(x_1) = f(x_2) \implies x_1 = x_2.$$



This function is not injective: $a \neq b$ but $f(a) = f(b) = u$.

A counterexample must give *two distinct inputs with the same output*. An unused output concerns surjectivity instead.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.1.2; Examples 8.1.8–8.1.11.

Two complete injectivity arguments

Translation: $q : \mathbb{R} \to \mathbb{R}$, $q(x) = x + 3$

Let $x_1, x_2 \in \mathbb{R}$ and assume $q(x_1) = q(x_2)$. Then

$$x_1 + 3 = x_2 + 3 \quad \Longrightarrow \quad x_1 = x_2$$

by subtracting 3. These inputs were arbitrary, so q is injective.

Squaring: $s : \mathbb{R} \to \mathbb{R}$, $s(x) = x^2$

Choose $x_1 = -1$ and $x_2 = 1$. We have

$$s(-1) = 1 = s(1), \qquad -1 \neq 1.$$

Thus s fails the defining implication and is not injective.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.1.3–8.1.4.

Surjectivity asks whether every target is reached

Definition

$f : X \to Y$ is *surjective* when $\forall y \in Y,\ \exists x \in X,\ f(x) = y$.

Example. Let $f : \mathbb{Q} \to \mathbb{Q}$ satisfy $f(x) = 3x + 2$. Given any $y \in \mathbb{Q}$, choose $x = (y - 2)/3 \in \mathbb{Q}$. Then

$$f(x) = 3\frac{y-2}{3} + 2 = y.$$

The witness depends on the arbitrary target y, so this proves surjectivity.

Failure. Squaring $\mathbb{R} \to \mathbb{R}$ misses -1, since $x^2 \geq 0$ for every real x. A failed search for a witness is insufficient; we proved no witness can exist.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.1.5; Examples 8.1.6–8.1.7.

From quantifiers to Lean proof steps

```
example : Function.Injective
    (fun x : Real => x + 3) := by
  intro x1 x2 h
  dsimp at h
  addarith [h]

example : Function.Surjective
    (fun x : Rat => 3 * x + 2) := by
  intro y
  use (y - 2) / 3
  dsimp
  ring
```

`intro` selects arbitrary inputs or a target; `use` supplies a preimage. For negative claims, unfold the property and use `push_neg`.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.1.3 and 8.1.6; illustrative textbook-environment code.

Practice 8.1: Give a proof or a specific obstruction

1. Is $x \mapsto 2x$ surjective as a function $\mathbb{R} \to \mathbb{R}$? As a function $\mathbb{Z} \to \mathbb{Z}$?
2. Let $X = \{a, b, c\}$ and $Y = \{u, v\}$, with

x	a	b	c
$f(x)$	u	v	u

   Is $f : X \to Y$ injective? Is it surjective?

For each positive answer, account for every relevant input or target. For each negative answer, state the exact witnesses.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 8.1.13; original finite-table exercise.

Practice 8.1: Solution ideas

1. Over $\mathbb{R}$, target y has preimage $y/2$, because $2(y/2) = y$. Over $\mathbb{Z}$, target 1 has no preimage: $2n$ is even for every $n \in \mathbb{Z}$.
2. The finite function is not injective: $a \neq c$ and $f(a) = f(c) = u$. It is surjective: u has preimage a, and v has preimage b. These are all the targets in Y.

Proof structure in Lean

Surjectivity over a finite type: `intro y`, then `cases y`, then a witness in each case. The book's `exhaust` handles the resulting concrete finite statements.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.1.9 and 8.1.11; Exercises 8.1.13.

8.2: Bijectivity combines two obligations

Definition

$f : X \to Y$ is *bijective* when it is both injective and surjective.

Example. Let $p : \mathbb{R} \to \mathbb{R}$, $p(x) = 2x - 5$.

1. **Injectivity.** If $p(x_1) = p(x_2)$, then $2x_1 - 5 = 2x_2 - 5$. Adding 5 and dividing by 2 gives $x_1 = x_2$.
2. **Surjectivity.** Given $y \in \mathbb{R}$, let $x = (y + 5)/2$. Then

$$p(x) = 2\frac{y+5}{2} - 5 = y.$$

Both parts hold with domain and codomain $\mathbb{R}$, so p is bijective.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.2.1; Example 8.2.2.

A bijection gives exactly one preimage

Theorem

For $f : X \to Y$, f is bijective if and only if

$$\forall y \in Y, \quad \exists! x \in X \text{ such that } f(x) = y.$$

Forward. Surjectivity gives a preimage x of y. If $f(x') = y$ also, then $f(x') = f(x)$; injectivity gives $x' = x$.

Backward. Existence gives surjectivity. To prove injectivity, assume $f(x_1) = f(x_2)$ and call their common output y. Both x_1 and x_2 are preimages of y, so uniqueness gives $x_1 = x_2$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.2.5.

Lean: Split the bijection proof

```
example : Function.Bijective
    (fun x : Real => 2 * x - 5) := by
  constructor
  case left =>
    intro x1 x2 h
    dsimp at h
    addarith [h]
  case right =>
    intro y
    use (y + 5) / 2
    dsimp
    ring
```

`constructor` separates injectivity from surjectivity. To refute bijectivity, it is enough to refute either part.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.2.2–8.2.4; illustrative textbook-environment code.

Practice 8.2: Two ways a bijection can fail

1. For $s : \mathbb{N} \to \mathbb{N}$, $s(n) = n + 1$, prove injectivity and disprove bijectivity.
2. For $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^2 + 2x$, disprove bijectivity using two inputs.

Plan before calculating

For the first function, look for a target below every output. For the second, factor $x^2 + 2x = x(x + 2)$ and compare its zeros.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.2.7; Exercises 8.2.8.

Practice 8.2: Solution ideas

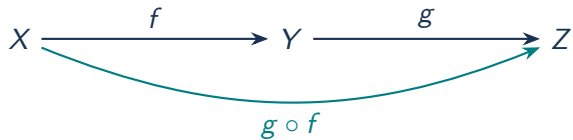
1. If $n_1 + 1 = n_2 + 1$, cancellation gives $n_1 = n_2$, so s is injective. But $s(n) = n + 1 > 0$ for all $n \in \mathbb{N}$. Thus 0 has no preimage, and s is not bijective.
2. We have $h(0) = 0$ and $h(-2) = 4 - 4 = 0$, whereas $0 \neq -2$. Thus h is not injective, which already disproves bijectivity.

Finite versus infinite domains

On the same finite type, an injective self-map is bijective (the book checks small examples). The successor function shows that the analogous statement for $\mathbb{N}$ is false.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.2.6–8.2.7; Exercises 8.2.8.

8.3: Composition follows the types



For $f : X \to Y$ and $g : Y \to Z$, define $(g \circ f)(x) = g(f(x))$. The identity $\mathrm{id}_X : X \to X$ satisfies $\mathrm{id}_X(x) = x$.

If $f(x) = x + 3$ and $g(x) = 2x$ on $\mathbb{R}$, then

$$(g \circ f)(x) = 2(x + 3) = 2x + 6, \qquad (f \circ g)(x) = 2x + 3.$$

Order matters. Equality of functions means equality at *every* input.

Source: Heather Macbeth, *The Mechanics of Proof*, Definitions 8.3.1 and 8.3.3; Example 8.3.2.

Lean: Extensionality changes the kind of goal

```
def f (x : Real) : Real := x + 3
def g (x : Real) : Real := 2 * x
def h (x : Real) : Real := 2 * x + 6

example : (fun x => g (f x)) = h := by
  ext x
  dsimp [f, g, h]
  ring
```

Before `ext x`	Equality of two functions
After `ext x`	Equality of their values at an arbitrary x
After `dsimp`	$2(x+3) = 2x+6$

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.3.2; illustrative textbook-environment code.

An inverse must undo the function in both directions

Definition

For $f : X \to Y$, a function $g : Y \to X$ is an *inverse* if

$$g \circ f = \mathrm{id}_X \qquad \text{and} \qquad f \circ g = \mathrm{id}_Y .$$

Equivalently: $g(f(x)) = x$ for every $x \in X$, and $f(g(y)) = y$ for every $y \in Y$.

For $s : \mathbb{R} \to \mathbb{R}$, $s(x) = 5 - x$, we obtain

$$s(s(x)) = 5 - (5 - x) = x \qquad (x \in \mathbb{R}).$$

Thus s is its own inverse. Since the two functions are the same here, the one calculation establishes both required identities.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 8.3.5; Example 8.3.4.

A function is bijective exactly when it has an inverse

Suppose $f : X \to Y$ is bijective. For each y, let $g(y)$ be its unique preimage. Then $f(g(y)) = y$. For any x,

$$f(g(f(x))) = f(x) \quad \Longrightarrow \quad g(f(x)) = x$$

by injectivity. Hence g is an inverse.

Suppose $g : Y \to X$ is an inverse. If $f(x_1) = f(x_2)$, then

$$x_1 = g(f(x_1)) = g(f(x_2)) = x_2.$$

For any $y \in Y$, $g(y) \in X$ is a preimage because $f(g(y)) = y$. Thus f is injective and surjective.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.3.7–8.3.9; Lean uses choose to select preimages.

Practice 8.3: Build a function proof from smaller ones

1. Let $u : \mathbb{R} \to \mathbb{R}$ satisfy $u(x) = 5x + 1$. Find an inverse and verify both compositions.
2. Let $f : X \to Y$ and $g : Y \to Z$ be injective. Prove $g \circ f$ is injective.
3. With the same types, assume instead that both functions are surjective. Prove $g \circ f$ is surjective.

For composition, write the intermediate value in Y explicitly. It is where the two hypotheses meet.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 8.3.10.

Practice 8.3: Solution ideas

1. Set $v(y) = (y-1)/5$. Then
$$v(u(x)) = \frac{5x+1-1}{5} = x, \qquad u(v(y)) = 5\frac{y-1}{5} + 1 = y.$$
2. If $g(f(x_1)) = g(f(x_2))$, injectivity of g gives $f(x_1) = f(x_2)$; injectivity of f then gives $x_1 = x_2$.
3. Given $z \in Z$, choose $y \in Y$ with $g(y) = z$, then choose $x \in X$ with $f(x) = y$. Consequently $g(f(x)) = g(y) = z$.

In Lean, the second proof applies two injectivity hypotheses; the third extracts two existential witnesses and then uses x.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 8.3.10.

8.4: Ordered pairs keep both coordinates

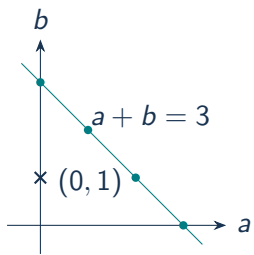
The product $X \times Y$ consists of pairs (x, y) with $x \in X$ and $y \in Y$.

$$(x_1, y_1) = (x_2, y_2) \iff x_1 = x_2 \ \wedge \ y_1 = y_2.$$

Let $q : \mathbb{Z} \to \mathbb{Z}^2$, $q(m) = (m + 1, 2 - m)$.

- Equal outputs give $m_1 + 1 = m_2 + 1$ from the first coordinate, hence $m_1 = m_2$.
- Every output (a, b) satisfies $a + b = 3$. Since $0 + 1 \neq 3$, $(0, 1)$ has no preimage.

So q is injective but not surjective.



Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.1.

Solve for an inverse, then verify it

Let $T : \mathbb{Z}^2 \to \mathbb{Z}^2$ satisfy $T(m, n) = (m + n, m + 2n)$. If $(a, b) = T(m, n)$, subtracting the two equations gives

$$n = b - a, \qquad m = a - n = 2a - b.$$

Thus define $U : \mathbb{Z}^2 \to \mathbb{Z}^2$ by $U(a, b) = (2a - b, b - a)$. Both components are integers. Now check:

$$\begin{aligned} U(T(m, n)) &= (2(m + n) - (m + 2n),\ (m + 2n) - (m + n)) \\ &= (m, n), \\ T(U(a, b)) &= ((2a - b) + (b - a),\ (2a - b) + 2(b - a)) \\ &= (a, b). \end{aligned}$$

U is an inverse, so T is bijective.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.2.

Lean: Function equality and coordinate equality

```
def T (p : Prod Int Int) : Prod Int Int :=
  (p.1 + p.2, p.1 + 2 * p.2)
def U (p : Prod Int Int) : Prod Int Int :=
  (2 * p.1 - p.2, p.2 - p.1)

example : And ((fun p => U (T p)) = id)
              ((fun p => T (U p)) = id) := by
  constructor <;> ext p <;> dsimp [T, U] <;> ring
```

- `Prod Int Int` is the ASCII spelling of $\mathbb{Z} \times \mathbb{Z}$.
- `p.1` and `p.2` access the two coordinates.
- `<;>` sends the following tactic to every generated goal.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.2; illustrative textbook-environment code.

Enumerating pairs of natural numbers

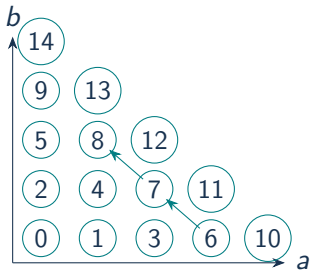
Theorem

There exists a bijection $\mathbb{N}^2 \to \mathbb{N}$.

Define $A_0 = 0$ and $A_{s+1} = A_s + s + 1$. The first values are $0, 1, 3, 6, 10, \ldots$.

$$p(a, b) = A_{a+b} + b.$$

On diagonal $a + b = s$, the second coordinate b runs from 0 through s. Thus this diagonal gets the labels $A_s, \ldots, A_s + s$.



Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.9.

Why the diagonal enumeration is a bijection

Diagonal s receives exactly the integer interval

$$I_s = \{A_s, A_s + 1, \ldots, A_s + s\} = \{A_s, \ldots, A_{s+1} - 1\}.$$

Injectivity. Different diagonals have disjoint intervals, since the intervals are consecutive and A_s strictly increases. Equal labels therefore have the same s. Within that diagonal, $A_s + b$ determines b, and then $a = s - b$.

Surjectivity. $I_0 = \{0\}$ and each interval starts just after the previous one ends. They cover $\mathbb{N}$: inductively, the successor of a covered number stays in its interval or is the first number of the next interval. If $k \in I_s$, take $b = k - A_s$ and $a = s - b$. Then $0 \leq b \leq s$ and $p(a, b) = k$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 8.4.9; alternative written proof of the same construction.

Practice 8.4: Check that the inverse stays in the domain

Consider $H(m, n) = (m + n, m - n)$.

1. Prove $H : \mathbb{R}^2 \to \mathbb{R}^2$ is bijective by constructing an inverse.
2. Decide whether the same rule gives a bijection $\mathbb{Z}^2 \to \mathbb{Z}^2$. Test the target $(0, 1)$.
3. Further practice: prove $(m, n) \mapsto 5m + 8n$ is surjective from $\mathbb{Z}^2$ to $\mathbb{Z}$. Use $5(-3) + 8(2) = 1$.

Solving equations over $\mathbb{R}$ does not ensure the resulting preimage has integer coordinates.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.4.3 and 8.4.6.

Practice 8.4: Solution ideas

1. Over $\mathbb{R}$, set $K(a,b) = ((a+b)/2, (a-b)/2)$. Then
$$K(H(m,n)) = (m,n), \qquad H(K(a,b)) = \left(\tfrac{a+b+a-b}{2}, \tfrac{a+b-a+b}{2}\right) = (a,b).$$
2. Over $\mathbb{Z}$, $H(m,n) = (0,1)$ would give $m+n=0$ and $m-n=1$. Adding gives $2m=1$, impossible for an integer m. Hence H is not surjective.
3. For any $z \in \mathbb{Z}$, the pair $(-3z, 2z)$ is an integer preimage: $5(-3z) + 8(2z) = z$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 8.4.3 and 8.4.6.

Chapter 8 review: Choose the obligation first

Goal	Useful first move
Injective	Assume equal outputs; recover equal inputs.
Not injective	Exhibit distinct inputs with equal outputs.
Surjective	Fix a target; construct a preimage in the domain.
Not surjective	Exhibit a target and rule out every preimage.
Bijective	Prove both properties, or verify an inverse.
Equal functions	Use an arbitrary input; in Lean, `ext`.
Equal pairs	Compare the corresponding coordinates.

Exit check. Explain exactly why division by 2 is harmless over $\mathbb{R}$ but can obstruct a claimed inverse over $\mathbb{Z}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Sections 8.1–8.4.

Reading and code context

Heather Macbeth, *The Mechanics of Proof*, Chapter 8:

- 8.1 Injectivity and surjectivity
- 8.2 Bijectivity
- 8.3 Composition of functions
- 8.4 Product types

`https://hrmacbeth.github.io/math2001/08_Functions.html`

These notes reorganize selected examples and supply new explanations and diagrams. Lean listings illustrate the textbook's teaching environment, including `addarith`, `numbers`, and `exhaust`. ASCII type names such as `Real` and `Prod Int Int` spell out the corresponding mathematical types.

Chapter 9
Sets

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What we will learn

- Translate membership, inclusion, and equality into logical statements.
- Prove statements about unions, intersections, and complements.
- Keep the ambient type explicit, including for empty and universal sets.
- Treat sets as inputs and outputs of functions.
- Prove that no function $X \to \mathcal{P}(X)$ can be surjective.

The organizing idea

A set in a type X is specified by a predicate on X. To ask whether x belongs to the set is to ask whether that predicate holds at x.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 9; Sections 9.1–9.3.

9.1: Membership evaluates a predicate

For a predicate P on X, write

$$S = \{x : X \mid P(x)\}, \qquad a \in S \iff P(a).$$

The notation $a \notin S$ means $\neg(a \in S)$.

A concrete set of integers

Let $S = \{n : \mathbb{Z} \mid n \leq 3\}$. Then $1 \in S$ because $1 \leq 3$, while $7 \notin S$ because $7 \not\leq 3$. Negative integers can also belong to S: for example, $-10 \leq 3$.

A set defined by parity

For $O = \{n : \mathbb{N} \mid n \text{ is odd}\}$, we have $10 \notin O$: $10 = 2 \cdot 5$ is even, and no natural number is both even and odd.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.1.1–9.1.2.

Inclusion is a universal implication

Definition

For sets U, V in the same type X,

$$U \subseteq V \quad \iff \quad \forall x \in X, \quad x \in U \implies x \in V.$$

Claim. $\{a : \mathbb{N} \mid 4 \mid a\} \subseteq \{b : \mathbb{N} \mid 2 \mid b\}$.

Proof. Choose an arbitrary $a \in \mathbb{N}$ with $4 \mid a$. There is a $k \in \mathbb{N}$ with $a = 4k$. Now

$$a = 4k = 2(2k).$$

The witness $2k \in \mathbb{N}$ proves $2 \mid a$. Therefore every element of the first set belongs to the second.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.1.3.

To refute inclusion, find an element that escapes

$$U \not\subseteq V \quad \Longleftrightarrow \quad \exists x \in X, \quad x \in U \ \wedge \ x \notin V.$$

For $U = \{x : \mathbb{R} \mid 0 \leq x^2\}$ and $V = \{x : \mathbb{R} \mid 0 \leq x\}$, the witness $x = -3$ works: $0 \leq (-3)^2 = 9$, but $-3 < 0$.

An element in just one of two sets also proves the sets unequal. For example,

$$\{n : \mathbb{N} \mid 4 \mid n\} \neq \{n : \mathbb{N} \mid 2 \mid n\}.$$

Indeed, $6 = 2 \cdot 3$, so 6 belongs to the second set. If $6 = 4k$ with $k \in \mathbb{N}$, then either $k \leq 1$ gives $4k \leq 4$, or $k \geq 2$ gives $4k \geq 8$. Thus 6 is absent from the first set.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.1.4 and 9.1.6.

Set equality means the same membership test

Extensionality

$U = V$ if and only if $\forall x \in X,\ x \in U \iff x \in V$. Equivalently, prove both $U \subseteq V$ and $V \subseteq U$.

Example. The sets

$$O = \{x : \mathbb{Z} \mid x \text{ is odd}\}, \qquad M = \{x : \mathbb{Z} \mid \exists k \in \mathbb{Z},\ x = 2k - 1\}$$

are equal. Fix $x \in \mathbb{Z}$.

- If $x \in O$, write $x = 2l + 1 = 2(l + 1) - 1$. The integer witness $k = l + 1$ gives $x \in M$.
- If $x \in M$, write $x = 2k - 1 = 2(k - 1) + 1$. The integer witness $l = k - 1$ shows that x is odd.

Both implications hold for every x, hence $O = M$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.1.5.

Finite sets turn membership into alternatives

For a fixed ambient type,

$$x \in \{a, b\} \quad \Longleftrightarrow \quad x = a \ \vee \ x = b.$$

Order and repetitions do not affect a set: $\{a, b, a\} = \{b, a\}$.

A complete solution set

We claim $\{x : \mathbb{R} \mid x^2 - x - 2 = 0\} = \{-1, 2\}$. For an arbitrary real x:

- If $x^2 - x - 2 = 0$, then $(x + 1)(x - 2) = 0$. Since $\mathbb{R}$ has no zero divisors, $x + 1 = 0$ or $x - 2 = 0$, giving $x = -1$ or $x = 2$.
- Conversely, substituting $x = -1$ gives $1 + 1 - 2 = 0$; substituting $x = 2$ gives $4 - 2 - 2 = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.1.8.

Lean: Strip off the set notation

```
example : Set.Subset
    {a : Nat | Exists (fun k : Nat => a = 4 * k)}
    {b : Nat | Exists (fun j : Nat => b = 2 * j)} := by
  intro a ha
  cases ha with
  | intro k hk =>
      use 2 * k
      calc a = 4 * k := hk
        _ = 2 * (2 * k) := by ring
```

Here `Exists` spells out the divisibility witness.

Membership	`dsimp` exposes the predicate.
Subset	`intro x hx` starts the implication.
Set equality	`ext x` creates a membership equivalence.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.1.1, 9.1.3, and 9.1.5; textbook-environment demonstration.

Practice 9.1: Do the sets contain exactly what is claimed?

1. Prove $\{n : \mathbb{N} \mid 20 \mid n\} \subseteq \{n : \mathbb{N} \mid 5 \mid n\}$. Is the reverse inclusion true?
2. Prove or disprove
$$\{t : \mathbb{R} \mid t^2 - 5t + 4 = 0\} = \{4\}.$$
3. Decide whether -3 belongs to $\{x : \mathbb{R} \mid \forall y \in \mathbb{R},\ x \leq y^2\}$.

State the logical form of each task before proving it. An equation satisfied by every listed element does not prove the list is complete.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.1.10.

Practice 9.1: Solution ideas

1. If $n = 20k$, then $n = 5(4k)$, so $5 \mid n$. The reverse inclusion fails at $n = 5$: it is a multiple of 5 but not of 20.
2. The equality is false. The witness $t = 1$ satisfies $1 - 5 + 4 = 0$, but $1 \notin \{4\}$. In fact, $(t-1)(t-4) = t^2 - 5t + 4$, so the full solution set is $\{1, 4\}$.
3. Yes. For every real y, $y^2 \geq 0$, hence $-3 \leq 0 \leq y^2$. The same input -3 works for *all* values quantified inside the predicate.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.1.10.

9.2: Set operations package familiar logical operations

Fix sets U, V in a type X and an element $x \in X$.

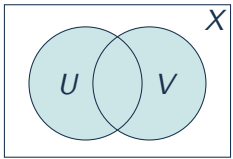
Set	Membership condition	Logical operation
$U \cup V$	$x \in U \vee x \in V$	or
$U \cap V$	$x \in U \wedge x \in V$	and
U^c	$x \notin U$	not
$\emptyset$	false	false
univ_X	true	true

The complement is relative to X: the complement of $\{0\}$ in $\mathbb{N}$ contains no negative numbers; its complement in $\mathbb{Z}$ does.

We often write X for its universal set on paper. Lean keeps the type `X` and the set `Set.univ` separate.

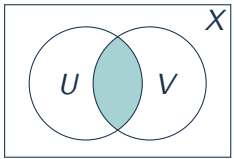
Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.2.1, 9.2.3, and 9.2.5–9.2.7.

Union accepts either membership; intersection requires both



$U \cup V$

To prove membership, establish at least one side. From membership, be ready for two cases.



$U \cap V$

To prove membership, establish both sides. From membership, obtain two usable facts.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.2.1 and 9.2.3; original diagrams.

A worked intersection proof uses both hypotheses

Claim.

$$\{-2, 3\} \cap \{x : \mathbb{Q} \mid x^2 = 9\} \subseteq \{x : \mathbb{Q} \mid 0 < x\}.$$

Proof. Fix $t \in \mathbb{Q}$ in the intersection. Then

$$(t = -2 \ \vee \ t = 3) \qquad \text{and} \qquad t^2 = 9.$$

- If $t = -2$, then $t^2 = 4$. Together with $t^2 = 9$, this gives $4 = 9$, a contradiction.
- If $t = 3$, then $0 < t$.

The first condition narrows the possibilities; the second rules out the unwanted one. Thus every element of the intersection is positive.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.2.3.

Empty sets and universal sets require different proofs

An empty intersection. Let $A = \{n : \mathbb{Z} \mid n \equiv 1 \pmod 5\}$ and $B = \{n : \mathbb{Z} \mid n \equiv 2 \pmod 5\}$. If $x \in A \cap B$, then $x = 1 + 5k = 2 + 5l$ for some integers k, l. Thus $1 = 5(k - l)$, impossible. Hence $A \cap B = \emptyset$.

A universal union. Let $C = \{x : \mathbb{R} \mid -1 < x\}$ and $D = \{x : \mathbb{R} \mid x < 1\}$. For an arbitrary $t \in \mathbb{R}$:

- if $t \leq 0$, then $t < 1$, so $t \in D$;
- if $t > 0$, then $-1 < t$, so $t \in C$.

Therefore every real number belongs to $C \cup D$, and $C \cup D = \mathbb{R}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 9.2.6–9.2.7.

Lean: Arithmetic facts versus propositional structure

```
example : Set.union ({1, 2} : Set Nat) {2, 4}
    = {1, 2, 4} := by
  ext n
  dsimp
  exhaust
```

After unfolding, the goal is the logical equivalence

$$((n = 1 \lor n = 2) \lor (n = 2 \lor n = 4)) \iff (n = 1 \lor n = 2 \lor n = 4).$$

The book's `exhaust` handles this propositional rearrangement. It does not replace a needed arithmetic argument such as proving two residue conditions incompatible.

For an intersection hypothesis, split the conjunction; for a union hypothesis, split into cases. These are the same proof patterns used before sets were introduced.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.2.2; Exercises 9.2.8.

Practice 9.2: Calculate, then prove a general identity

1. In the ambient type $\mathbb{N}$, simplify

$$\{0,1,2,3,4\} \cap \{0,2,4,6,8\}, \qquad \{3,4,5\}^c \cap \{1,3,5,7,9\}.$$

2. For arbitrary sets U, V in the same type X, prove

$$(U \cup V)^c = U^c \cap V^c.$$

Start with an arbitrary $x \in X$ and translate each side into logic.

The picture can suggest an answer. A membership proof establishes it for every element.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.2.8; original general-identity exercise.

Practice 9.2: Solution ideas

The explicit intersections are $\{0, 2, 4\}$ and $\{1, 7, 9\}$. In the second expression, remove 3 and 5 from the listed set; 4 was absent already.

For the identity, fix $x \in X$.

- Suppose $x \notin U \cup V$. If $x \in U$, it would belong to the union; similarly for V. Hence $x \notin U$ and $x \notin V$, so $x \in U^c \cap V^c$.
- Suppose $x \in U^c \cap V^c$. Then $x \notin U$ and $x \notin V$. Either case of $x \in U \cup V$ contradicts one of these facts. Thus $x \notin U \cup V$.

The membership conditions are equivalent for every x, so extensionality gives the set equality.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 9.2 definitions; solutions to the preceding practice.

9.3: A set can itself be an element

The type of all sets in X is written $\mathcal{P}(X)$ on paper and `Set X` in Lean.

Object	Type
3	$\mathbb{N}$
$\{3, 4\}$	$\mathcal{P}(\mathbb{N})$
$\{\{3, 4\}, \{4, 5, 6\}\}$	$\mathcal{P}(\mathcal{P}(\mathbb{N}))$
$\mathcal{F} = \{S : \mathcal{P}(\mathbb{N}) \mid 3 \in S\}$	$\mathcal{P}(\mathcal{P}(\mathbb{N}))$

For the set E of even natural numbers, $E \notin \mathcal{F}$ because $3 \notin E$. However, $\{3, 4\} \in \mathcal{F}$ because $3 \in \{3, 4\}$.

Membership $S \in \mathcal{F}$ and inclusion $S \subseteq T$ are different statements about different types.

Source: Heather Macbeth, *The Mechanics of Proof*, Definition 9.3.1; Example 9.3.2.

A function on sets can lose information

Define $p : \mathcal{P}(\mathbb{N}) \to \mathcal{P}(\mathbb{N})$ by

$$p(S) = \{n : \mathbb{N} \mid n + 1 \in S\}.$$

For instance, $p(\{0, 2, 4\}) = \{1, 3\}$.

Claim: p is not injective

Take $S = \{0\}$ and $T = \emptyset$. Since $n + 1 > 0$ for every $n \in \mathbb{N}$, no n satisfies $n + 1 \in S$. Thus $p(S) = \emptyset = p(T)$. But $S \neq T$, because $0 \in S$ and $0 \notin T$.

The output never tests whether $0 \in S$; all tested inputs have the form $n + 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.3.

The same rule over integers preserves information

Define $q : \mathcal{P}(\mathbb{Z}) \to \mathcal{P}(\mathbb{Z})$ by $q(S) = \{n : \mathbb{Z} \mid n + 1 \in S\}$.

Claim. q is injective. Suppose $q(S) = q(T)$. To prove $S = T$, fix an arbitrary $k \in \mathbb{Z}$. Then $k - 1$ is also an integer, and

$$\begin{aligned} k \in S &\iff (k-1) + 1 \in S \\ &\iff k - 1 \in q(S) \\ &\iff k - 1 \in q(T) \\ &\iff (k-1) + 1 \in T \\ &\iff k \in T. \end{aligned}$$

Extensionality gives $S = T$. The ability to use $k - 1$ is essential.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.4.

Cantor's construction: A set that differs on the diagonal

Theorem

For every type X, no function $f : X \to \mathcal{P}(X)$ is surjective.

Define $D = \{x : X \mid x \notin f(x)\}$. The membership of x in D is the opposite of its membership in $f(x)$.

	a	b	c	d
$f(a)$	**1**	0	1	0
$f(b)$	1	**0**	0	1
$f(c)$	0	1	**1**	1
$f(d)$	1	0	0	**0**
D	0	1	0	1

In this illustrative finite table, 1 means membership and 0 means nonmembership.

D differs from row $f(a)$ at column a, from row $f(b)$ at column b, and so on. The definition works without enumerating X.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.5; original membership matrix.

Cantor's proof: The proposed preimage contradicts itself

Assume $f : X \to \mathcal{P}(X)$ is surjective and set $D = \{x : X \mid x \notin f(x)\}$. Surjectivity supplies $d \in X$ with $f(d) = D$. Consequently,

$$d \in D \quad \iff \quad d \notin f(d) \quad \iff \quad d \notin D.$$

- If $d \in D$, the defining condition gives $d \notin f(d) = D$, a contradiction.
- If $d \notin D$, then $d \notin f(d)$ because $f(d) = D$. By the definition of D, this says $d \in D$, again a contradiction.

Thus the assumption of surjectivity is impossible.

The obstruction applies to every proposed function f; the diagonal set is built from that very function.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.5.

Lean: The diagonal set is a predicate

```
example {X : Type} (f : X -> Set X) : Not (Function.Surjective f) := by
  intro hf
  let D : Set X := fun x => Not ((f x) x)
  cases hf D with
  | intro d hd =>
      by_cases h : D d
      case pos =>
        have hn : Not ((f d) d) := h
        rw [hd] at hn
        exact hn h
      case neg =>
        have hn : Not (Not ((f d) d)) := h
        rw [hd] at hn
        exact hn h
```

Here `D d` means $d \in D$, since a set is represented by a predicate.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 9.3.5; textbook-environment demonstration.

Practice 9.3: Sets as function values and sequence terms

1. Define $r : \mathcal{P}(\mathbb{N}) \to \mathcal{P}(\mathbb{N})$ by $r(S) = S \cup \{3\}$. Prove that r is not injective.
2. Define a sequence of sets of integers by

$$U_0 = \mathbb{Z}, \qquad U_{n+1} = \{x : \mathbb{Z} \mid \exists y \in U_n,\ x = 2y\}.$$

   Prove by induction that $U_n = \{x : \mathbb{Z} \mid 2^n \mid x\}$ for every $n \in \mathbb{N}$.

In the second task, the induction hypothesis is an equality of sets. Use it to replace a membership condition, then translate divisibility into a witness equation.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.3.6.

Practice 9.3: Solution ideas

The union function. The distinct inputs $\emptyset$ and $\{3\}$ both map to $\{3\}$. They are distinct because only the second contains 3.

The sequence. For $n = 0$, $2^0 = 1$ divides every integer, so the formula gives $U_0 = \mathbb{Z}$. Assume the formula for U_n and fix $x \in \mathbb{Z}$.

- If $x \in U_{n+1}$, write $x = 2y$ with $y \in U_n$. The induction hypothesis gives $y = 2^n k$ for some $k \in \mathbb{Z}$. Then $x = 2^{n+1} k$.
- If $x = 2^{n+1} k$, set $y = 2^n k$. Then $y \in U_n$ by the induction hypothesis and $x = 2y$, so $x \in U_{n+1}$.

These implications prove the required set equality at $n + 1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 9.3.6.

Chapter 9 review: Expose the underlying logic

Set statement	Proof obligation
$x \in \{t \mid P(t)\}$	Prove $P(x)$.
$U \subseteq V$	Start with arbitrary $x \in U$; prove $x \in V$.
$U = V$	Prove both directions of membership.
$U \neq V$	Find an element in exactly one set.
$U = \emptyset$	Rule out an arbitrary proposed member.
$U = \mathrm{univ}_X$	Show every $x \in X$ belongs.
$S \in \mathcal{F}$	Apply the family's predicate to the set S.

Exit check. For $\mathcal{F} = \{S : \mathcal{P}(\mathbb{N}) \mid 0 \in S\}$, explain why $\{0\} \in \mathcal{F}$ is well-typed and why $0 \in \mathcal{F}$ is not.

Source: Heather Macbeth, *The Mechanics of Proof*, Sections 9.1–9.3.

Reading and code context

Heather Macbeth, *The Mechanics of Proof*, Chapter 9:

- 9.1 Introduction
- 9.2 Set operations
- 9.3 The type of sets

https://hrmacbeth.github.io/math2001/09_Sets.html

These notes use original explanations, diagrams, and practice solutions around the cited material. Lean examples use the textbook's teaching environment. In ASCII listings, `Set.Subset` names inclusion, `Set.union` names union, and `Exists` introduces an existential predicate.

Chapter 10
Relations

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What we will learn

- Test reflexivity, symmetry, antisymmetry, and transitivity separately.
- Give the exact witnesses needed to refute each property.
- Prove that a relation is an equivalence relation.
- Define equivalence classes and prove why classes agree or are disjoint.
- Recognize equivalence relations on numbers, pairs, sets, and types.

A relation is a predicate with two inputs

A relation r on X assigns a proposition $r(x, y)$ to each pair $x, y \in X$. We write $x \sim y$ for $r(x, y)$. In Lean its type is `X -> X -> Prop`.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 10 introduction; Sections 10.1–10.2.

10.1: Four properties, four different statements

Fix a relation $\sim$ on a type X. All displayed variables range over X.

Property	Definition
Reflexive	$\forall x, \quad x \sim x.$
Symmetric	$\forall x, y, \quad x \sim y \implies y \sim x.$
Antisymmetric	$\forall x, y, \quad (x \sim y \ \wedge \ y \sim x) \implies x = y.$
Transitive	$\forall x, y, z, \quad (x \sim y \ \wedge \ y \sim z) \implies x \sim z.$

Antisymmetry permits $x \sim x$. It prohibits two-way relatedness between *distinct* elements.

Symmetry and antisymmetry are compatible: equality on $\mathbb{R}$ has both properties.

Source: Heather Macbeth, *The Mechanics of Proof*, Definitions in Example 10.1.1; Example 10.1.2.

Counterexamples must match the negated quantifiers

To disprove	Exhibit
Reflexivity	An x for which $x \not\sim x$.
Symmetry	x, y such that $x \sim y$ but $y \not\sim x$.
Antisymmetry	Distinct x, y with both $x \sim y$ and $y \sim x$.
Transitivity	x, y, z with $x \sim y$ and $y \sim z$, but $x \not\sim z$.

Lean pattern. Unfold the property, use `push_neg`, and provide the witnesses with `use`. Then verify each required fact.

For transitivity, just finding an unrelated pair proves nothing. The two premises must hold for the same middle element y.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 10.1.1–10.1.4.

Divisibility on the natural numbers

Let $a \sim b$ mean $a \mid b$, with $a, b \in \mathbb{N}$. Recall: $a \mid b$ means $b = ak$ for some $k \in \mathbb{N}$.

- **Reflexive.** For every a, $a = a \cdot 1$, so $a \mid a$.
- **Not symmetric.** $1 \mid 2$, but $2 \nmid 1$. Indeed, every natural multiple of 2 is either 0 or at least 2.
- **Transitive.** Assume $a \mid b$ and $b \mid c$. Write $b = ak$ and $c = bl$ with $k, l \in \mathbb{N}$. Then

$$c = bl = (ak)l = a(kl).$$

  Since $kl \in \mathbb{N}$, this is a witness for $a \mid c$.

Antisymmetry also holds, but requires attention to the value 0.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.1.

Antisymmetry of divisibility: Handle zero before comparing sizes

Suppose $a, b \in \mathbb{N}$ with $a \mid b$ and $b \mid a$. We must prove $a = b$.

1. If $a = 0$, the equation $b = ak$ from $a \mid b$ gives $b = 0$. If $b = 0$, the equation $a = bl$ gives $a = 0$.
2. Otherwise $a, b > 0$. Write $b = ak$ and $a = bl$ with $k, l \in \mathbb{N}$. The positivity of b forces $k \geq 1$, and the positivity of a forces $l \geq 1$.
3. Therefore $b = ak \geq a$ and $a = bl \geq b$, so $a = b$.

The domain matters

On $\mathbb{Z}$, divisibility is not antisymmetric: $1 \mid -1$ and $-1 \mid 1$, but $1 \neq -1$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.1; integer contrast added for comparison.

Lean: A transitivity proof combines two witnesses

```
def divides (a b : Nat) : Prop :=
  Exists (fun k : Nat => b = a * k)

example : Transitive divides := by
  intro a b c hab hbc
  cases hab with
  | intro k hk =>
      cases hbc with
      | intro l hl =>
          use k * l
          calc c = b * l := hl
            _ = (a * k) * l := by rw [hk]
            _ = a * (k * l) := by ring
```

The middle value b connects the equations. The new witness is the product kl.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.1; textbook-environment demonstration.

Being close is reflexive and symmetric, but need not be transitive

On $\mathbb{R}$, define $x \sim y$ by $(x-y)^2 \leq 1$.

- **Reflexive:** $(x-x)^2 = 0 \leq 1$.
- **Symmetric:** $(y-x)^2 = (x-y)^2$, so the inequality survives reversal.
- **Not antisymmetric:** $0 \sim 1$ and $1 \sim 0$ since both squared differences are 1; yet $0 \neq 1$.
- **Not transitive:** $0 \sim 1$ and $1 \sim 2$, but $(0-2)^2 = 4 > 1$, so $0 \not\sim 2$.

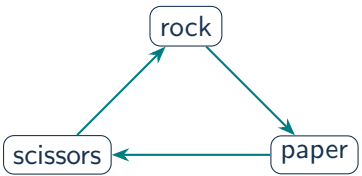


Each short step is allowed; the combined step is not.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.3; new counterexample choices and diagram.

A finite directed cycle is antisymmetric

Let $x \prec y$ mean "x loses to y" for rock, paper, and scissors. The diagram lists all related pairs.



- Not reflexive: there is no loop at rock.
- Not symmetric: rock $\prec$ paper, but paper $\not\prec$ rock.
- Antisymmetric: no pair has arrows in both directions, so its two-premise implication always holds.
- Not transitive: rock $\prec$ paper and paper $\prec$ scissors, but rock $\not\prec$ scissors.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.1.4; original drawing.

Practice 10.1: Prove or refute each property

For each relation, decide whether it is reflexive, symmetric, antisymmetric, and transitive.

1. On $\mathcal{P}(\mathbb{N})$, define $U \sim V$ by $U \subseteq V$. Your variables are now sets of natural numbers.
2. On $\mathbb{R}^2$, define

$$(x_1, y_1) \preceq (x_2, y_2) \quad \Longleftrightarrow \quad x_1 \leq x_2 \ \wedge \ y_1 \leq y_2.$$

   Your variables are now ordered pairs of real numbers.

Each positive answer needs a universal proof; each negative answer needs a suitable counterexample.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.1.5.

Practice 10.1: Inclusion on sets of natural numbers

- **Reflexive.** $U \subseteq U$: an arbitrary element of U already belongs to U.
- **Not symmetric.** $\emptyset \subseteq \{0\}$, but $\{0\} \not\subseteq \emptyset$ because $0 \notin \emptyset$.
- **Antisymmetric.** If $U \subseteq V$ and $V \subseteq U$, then for every $n \in \mathbb{N}$, $n \in U \iff n \in V$. Set extensionality gives $U = V$.
- **Transitive.** If $U \subseteq V$ and $V \subseteq W$, take $n \in U$. The first inclusion gives $n \in V$, and the second gives $n \in W$. Thus $U \subseteq W$.

The chapter on sets already supplied the proof of antisymmetry; here we recognize its role as a relation property.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.1.5; set extensionality from Section 9.1.

Practice 10.1: Compare the coordinates separately

For $\preceq$ on $\mathbb{R}^2$:

- **Reflexive.** $x \leq x$ and $y \leq y$, so $(x, y) \preceq (x, y)$.
- **Not symmetric.** $(0, 0) \preceq (1, 1)$, but the reverse would require $1 \leq 0$.
- **Antisymmetric.** Relations in both directions give $x_1 \leq x_2 \leq x_1$ and $y_1 \leq y_2 \leq y_1$. Thus $x_1 = x_2$ and $y_1 = y_2$, so the pairs are equal.
- **Transitive.** From $(x_1, y_1) \preceq (x_2, y_2)$ and $(x_2, y_2) \preceq (x_3, y_3)$, obtain $x_1 \leq x_2 \leq x_3$ and $y_1 \leq y_2 \leq y_3$. Hence $(x_1, y_1) \preceq (x_3, y_3)$.

Not every two points are comparable: $(0, 1)$ and $(1, 0)$ provide an example.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.1.5.

10.2: Equivalence relations formalize a notion of sameness

Definition

A relation is an *equivalence relation* if it is reflexive, symmetric, and transitive.

Fix any $n \in \mathbb{Z}$. On $\mathbb{Z}$, let $x \sim y$ mean $x \equiv y \pmod{n}$, equivalently $y - x = nk$ for some $k \in \mathbb{Z}$.

- **Reflexive:** $x - x = n \cdot 0$.
- **Symmetric:** if $y - x = nk$, then $x - y = n(-k)$.
- **Transitive:** if $y - x = nk$ and $z - y = nl$, then $z - x = (z - y) + (y - x) = n(l + k)$.

The argument needs no positivity assumption on n. When $n = 0$, the relation is equality, because $0 \mid (y - x)$ holds exactly when $y - x = 0$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.1.

Equivalence classes collect the elements related to one element

For a relation on X and $a \in X$, define

$$[a] = \{b : X \mid a \sim b\}.$$

For congruence modulo 3 on $\mathbb{Z}$, there are three classes:

$$[0] = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$$

$$[1] = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$$

$$[2] = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$$

Two integers are related exactly when they lie in the same row. Every integer is in its own class because the relation is reflexive.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 10.2.1 and 10.2.3.

Equal squares give another equivalence relation

On $\mathbb{Z}$, define $a \sim b$ by $a^2 = b^2$.

- **Reflexive:** $a^2 = a^2$ for every integer a.
- **Symmetric:** $a^2 = b^2$ implies $b^2 = a^2$.
- **Transitive:** $a^2 = b^2$ and $b^2 = c^2$ imply $a^2 = c^2$.

What are its classes? Since

$$a^2 = b^2 \quad \Longleftrightarrow \quad (a - b)(a + b) = 0,$$

we have $b = a$ or $b = -a$. Thus $[0] = \{0\}$ and $[a] = \{a, -a\}$ for $a \neq 0$.

This relation is not antisymmetric: $1 \sim -1$ and $-1 \sim 1$, yet $1 \neq -1$. Equivalence does not require antisymmetry.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.2.

Lean: Reuse the corresponding property of equality

```
def sameSquare (x y : Int) : Prop := x ^ 2 = y ^ 2

example : Reflexive sameSquare := by
  intro x
  rfl

example : Symmetric sameSquare := by
  intro x y h
  exact h.symm

example : Transitive sameSquare := by
  intro x y z hxy hyz
  exact hxy.trans hyz
```

Once the relation unfolds, reflexivity, symmetry, and transitivity become the corresponding facts about equality of squares.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.2; textbook-environment demonstration.

Related elements have equal classes

Theorem with its precise hypotheses

If $\sim$ is symmetric and transitive on X, and $a_1 \sim a_2$, then $[a_1] = [a_2]$.

Fix $b \in X$. We prove $a_1 \sim b \iff a_2 \sim b$.

- If $a_1 \sim b$, symmetry gives $a_2 \sim a_1$. Transitivity along $a_2 \sim a_1 \sim b$ gives $a_2 \sim b$.
- If $a_2 \sim b$, transitivity along $a_1 \sim a_2 \sim b$ gives $a_1 \sim b$.

Extensionality now gives the equality of sets. Reflexivity was not needed for this theorem.

Separately, if the relation is reflexive, then $a \in [a]$ for every a, because that membership statement is exactly $a \sim a$.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.3.

Why equivalence classes form a partition

Assume $\sim$ is an equivalence relation on X.

- **The classes cover X.** Every $a \in X$ belongs to $[a]$ by reflexivity.
- **Overlapping classes are equal.** Suppose $c \in [a] \cap [b]$. Then $a \sim c$ and $b \sim c$. Symmetry gives $c \sim b$, and transitivity gives $a \sim b$. The preceding theorem yields $[a] = [b]$.

Thus any two classes are either equal or disjoint. The distinct classes split X into nonempty, nonoverlapping pieces: a *partition*.

For equality on $\mathbb{Z}$, each class is a singleton: $[a] = \{a\}$. For equal squares on $\mathbb{Z}$, most classes contain two integers instead.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.3 and Example 10.2.4; partition corollary derived here.

Pairs can represent the same rational value

On $\mathbb{Z} \times \mathbb{N}$, define

$$(a, b) \sim (c, d) \quad \iff \quad a(d+1) = c(b+1).$$

The products are taken in $\mathbb{Z}$; the natural numbers b, d are viewed as integers. Since $b+1, d+1 > 0$, this condition means

$$\frac{a}{b+1} = \frac{c}{d+1} \qquad \text{in } \mathbb{Q}.$$

For example, $(1, 1) \sim (2, 3)$ because $1 \cdot 4 = 2 \cdot 2$.

- Reflexivity follows from $a(b+1) = a(b+1)$.
- Symmetry follows by reversing the equality $a(d+1) = c(b+1)$.

Transitivity requires a calculation and cancellation of a nonzero factor.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.5.

Transitivity for the pair representation

Assume $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Set $B = b + 1$, $D = d + 1$, and $F = f + 1$, as integers. Then

$$aD = cB, \qquad cF = eD, \qquad D > 0.$$

We want $aF = eB$. First calculate

$$\begin{aligned} D(aF) &= (aD)F \\ &= (cB)F \\ &= (cF)B \\ &= (eD)B \\ &= D(eB). \end{aligned}$$

Since $D \neq 0$, cancellation gives $aF = eB$, as required. In the book's Lean proof, `ring` rearranges products, `rw` uses the hypotheses, and `cancel D at this` performs the final cancellation.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.5.

Having a bijection is an equivalence relation on types

For level-0 types X, Y (Lean: `Type`), define

$$X \sim Y \quad \Longleftrightarrow \quad \text{there exists a bijective function } f : X \to Y.$$

- **Reflexive.** The identity $\mathrm{id}_X : X \to X$ is its own inverse, so it is bijective.
- **Symmetric.** If $f : X \to Y$ is bijective, it has an inverse $g : Y \to X$. The two inverse identities also show that f is an inverse of g; hence g is bijective.
- **Transitive.** Given bijections $f : X \to Y$ and $g : Y \to Z$, their composition $g \circ f : X \to Z$ is injective and surjective by the composition proofs in Chapter 8.

The relation records that two types can be matched exactly, even when their elements look different.

Source: Heather Macbeth, *The Mechanics of Proof*, Example 10.2.6; Section 8.3.

Practice 10.2: Prove equivalence and describe the classes

1. On $\mathbb{Z}$, define $a \sim b$ when there are positive integers m, n such that $am = bn$. Prove this is an equivalence relation and determine its classes.
2. On $\mathbb{N}^2$, define
$$(a, b) \approx (c, d) \quad \iff \quad a + d = b + c.$$
Prove this is an equivalence relation. Describe its classes using the integer difference $a - b$.

In the first problem, positivity is part of the definition. In the second, interpret differences in $\mathbb{Z}$, not as truncated subtraction in $\mathbb{N}$.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.2.7.

Practice 10.2: Positive multiples and the three sign classes

- **Reflexive:** use $m = n = 1$.
- **Symmetric:** if $am = bn$ with $m, n > 0$, reverse the equation and swap the witnesses.
- **Transitive:** suppose $am = bn$ and $bp = cq$, with all four witnesses positive. Then $a(mp) = b(np) = c(qn)$; both mp and qn are positive.

There are exactly three classes: the positive integers, $\{0\}$, and the negative integers.

- Positive multiplication preserves sign, so related integers have the same sign, including being zero together.
- If $a, b > 0$, choose $m = b, n = a$. If $a, b < 0$, choose $m = -b, n = -a$. These positive witnesses show that any two integers in either sign group are related.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.2.7, first problem.

Practice 10.2: Equal differences of natural-number pairs

For $(a, b), (c, d) \in \mathbb{N}^2$, the equation $a + d = b + c$ is equivalent to

$$a - b = c - d \qquad \text{when evaluated in } \mathbb{Z}.$$

- Reflexivity is $a + b = b + a$.
- Symmetry follows by reversing the equality of integer differences.
- Transitivity follows from $a - b = c - d$ and $c - d = e - f$, which give $a - b = e - f$, and hence $a + f = b + e$.

For each $k \in \mathbb{Z}$, one class is

$$C_k = \{(a, b) \in \mathbb{N}^2 \mid a - b = k \text{ in } \mathbb{Z}\}.$$

Each class is nonempty: use $(k, 0)$ if $k \geq 0$, or $(0, -k)$ if $k < 0$. Geometrically these classes lie on parallel diagonals in the natural-number grid.

Source: Heather Macbeth, *The Mechanics of Proof*, Exercises 10.2.7, second problem.

Chapter 10 review: Classify the relation before using it

Relation and ambient type	Refl.	Symm.	Antisymm.	Trans.
$=$ on $\mathbb{R}$	yes	yes	yes	yes
$\mid$ on $\mathbb{N}$	yes	no	yes	yes
$(x-y)^2 \leq 1$ on $\mathbb{R}$	yes	yes	no	no
Loses-to on three hands	no	no	yes	no
$a^2 = b^2$ on $\mathbb{Z}$	yes	yes	no	yes

An equivalence relation needs the first, second, and fourth properties. Its equivalence classes cover the type, and overlapping classes are equal.

Exit check. If an equivalence relation is also antisymmetric, prove it is equality: $a \sim b$ gives $b \sim a$ by symmetry, hence $a = b$; reflexivity gives the converse.

Source: Heather Macbeth, *The Mechanics of Proof*, Examples 10.1.1–10.1.4; Section 10.2.

Reading and code context

Heather Macbeth, *The Mechanics of Proof*, Chapter 10:

- 10.1 Reflexive, symmetric, antisymmetric, transitive
- 10.2 Equivalence relations

https://hrmacbeth.github.io/math2001/10_Relations.html

These notes include original diagrams, worked practice solutions, and the derived partition corollary. Lean examples use the textbook's teaching environment; its relation-property names are `Reflexive`, `Symmetric`, `AntiSymmetric`, and `Transitive`. The spelling of antisymmetry follows that source environment.